Week 8 — Simple regression II: inference & prediction

Business Forecasting

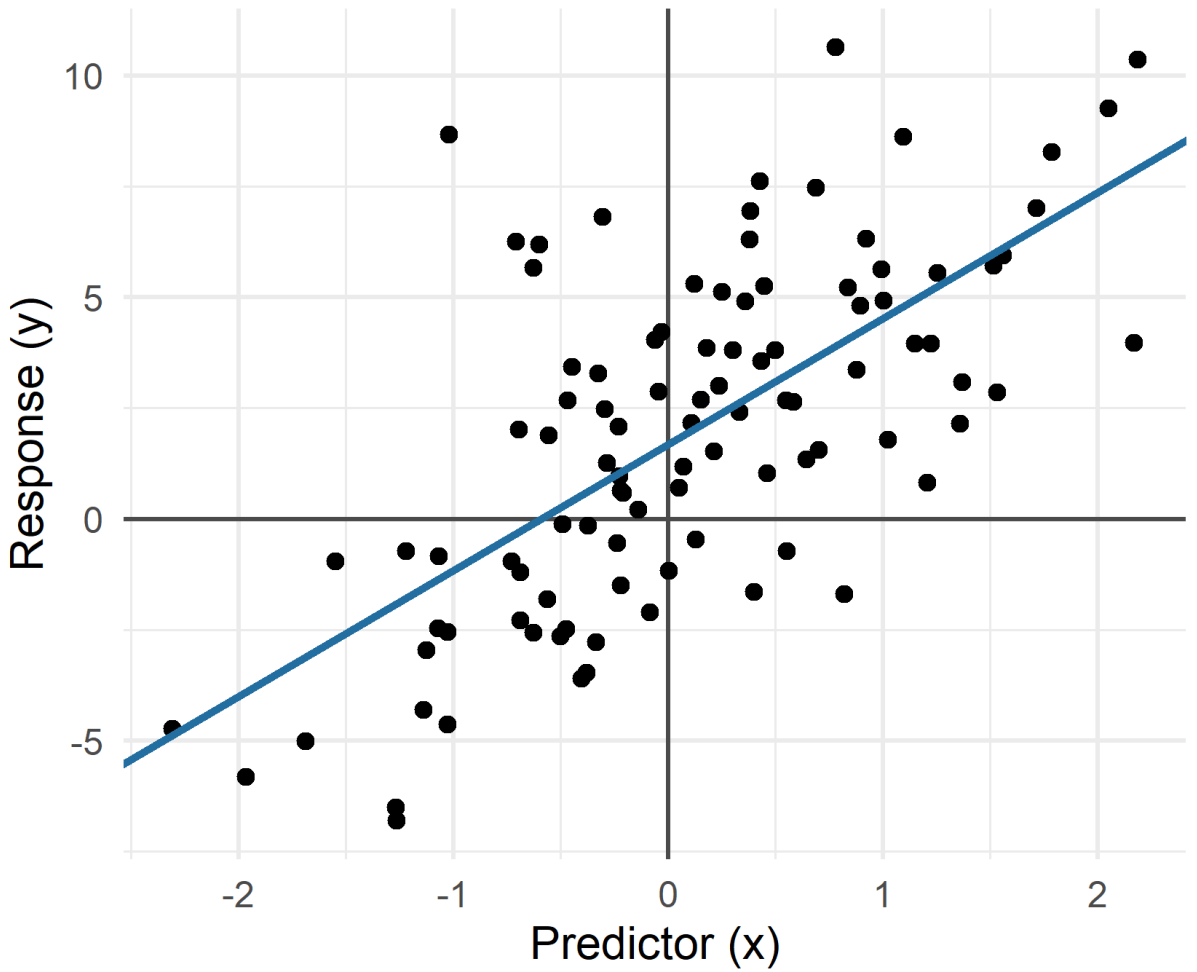
My code (.R) Notes

1 · The slope is a random variable

We only have samples, and yet we want to learn something about the population.

Uncertainty in the estimate

A population we can see: 100 points where we know the true line.

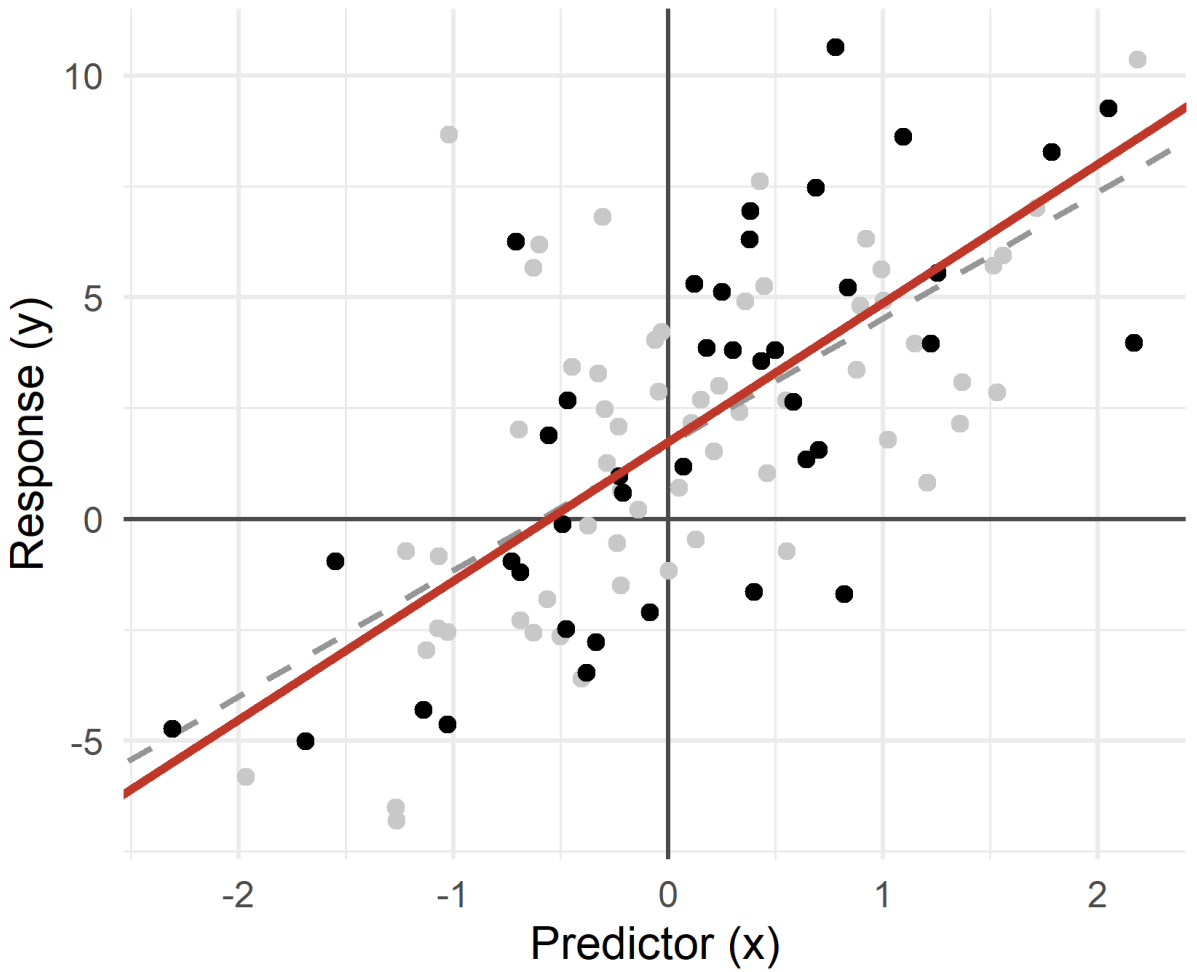


Population regression

$$y_i = 1.69 + 2.84\, x_i + u_i$$

Uncertainty in the estimate

A random sample of 40 from the same population.



Population regression

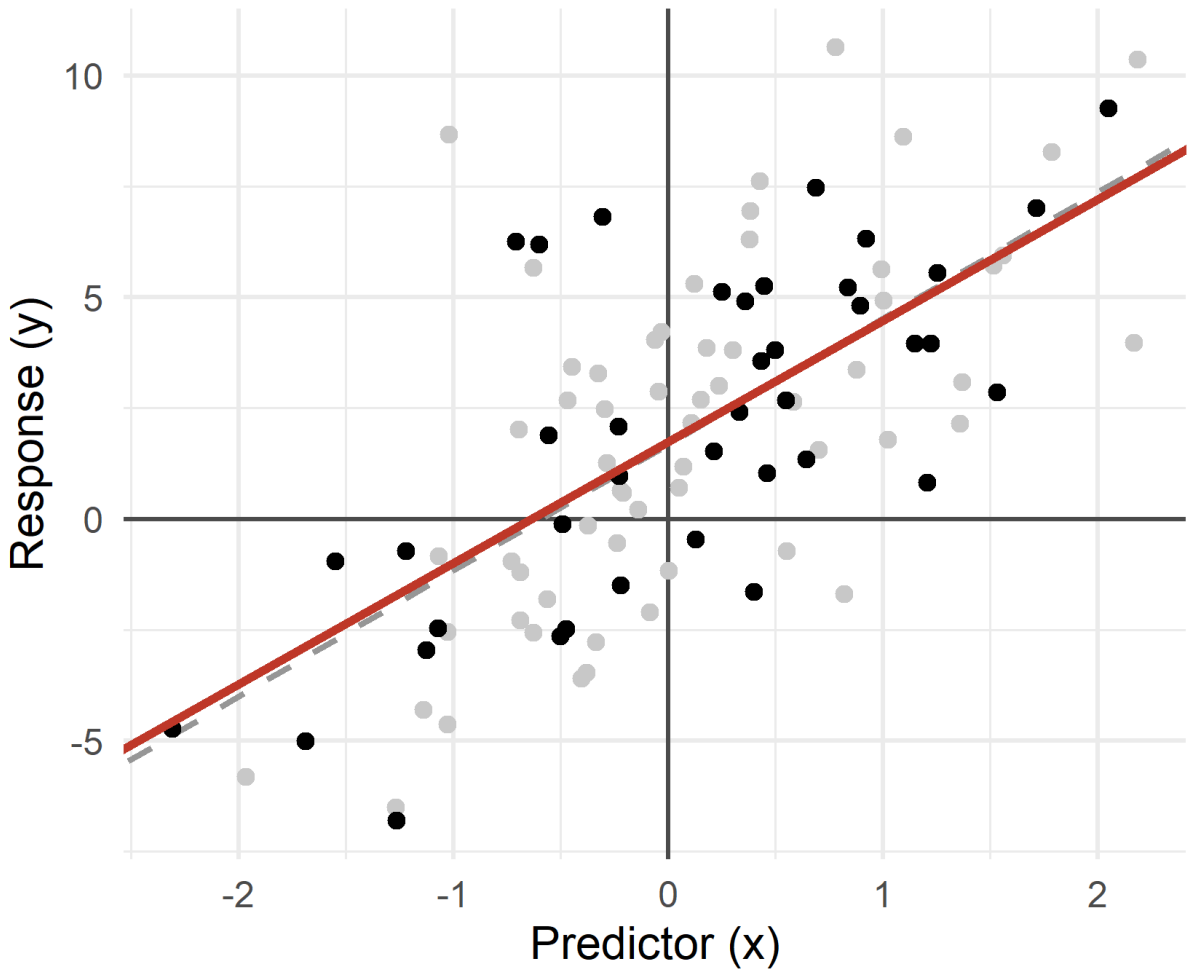
$$y_i = 1.69 + 2.84\, x_i + u_i$$

Sample 1 estimate

$$\hat{y}_i = 1.75 + 3.13\, x_i$$

Uncertainty in the estimate

A random sample of 40 from the same population.



Population regression

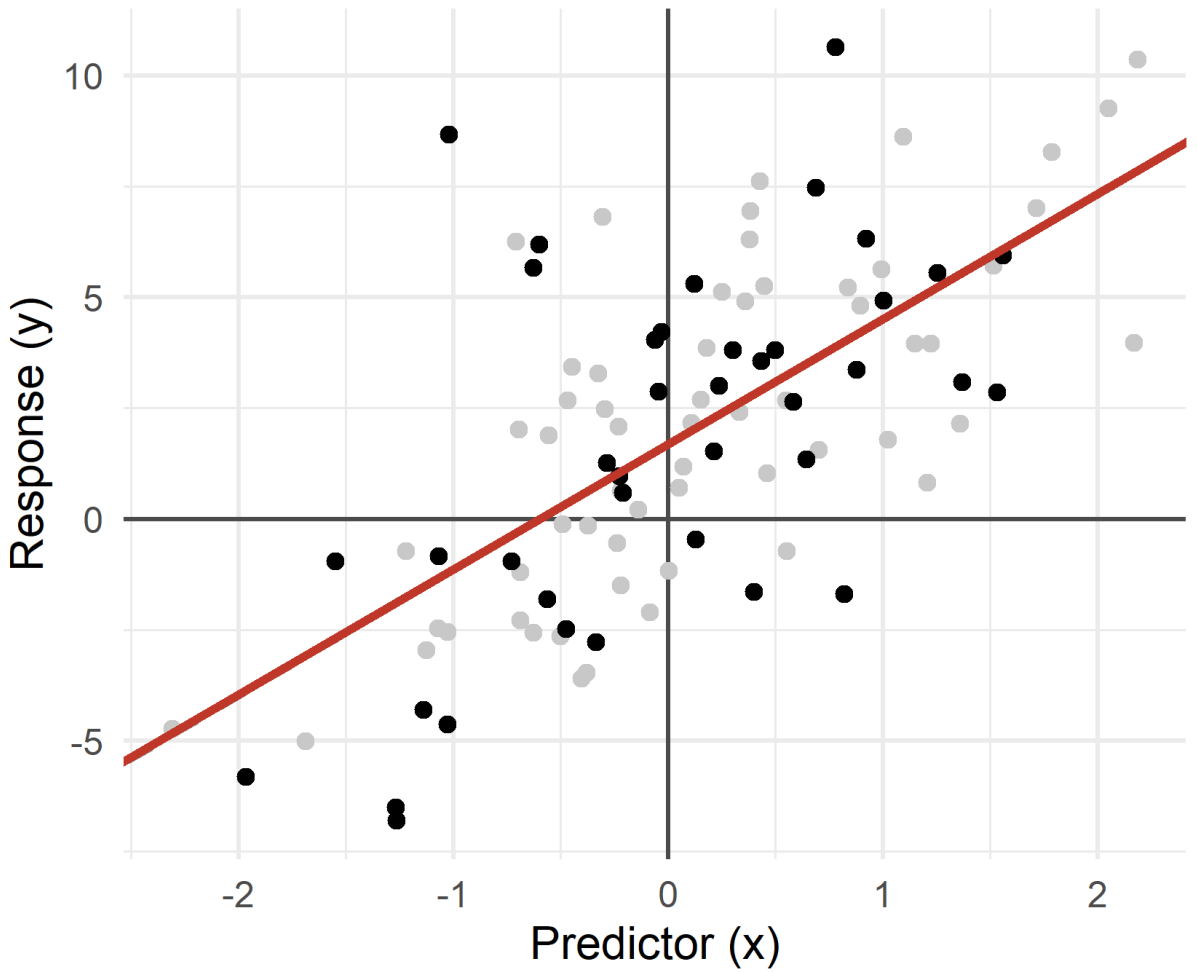
$$y_i = 1.69 + 2.84\, x_i + u_i$$

Sample 2 estimate

$$\hat{y}_i = 1.76 + 2.73\, x_i$$

Uncertainty in the estimate

A random sample of 40 from the same population.



Population regression

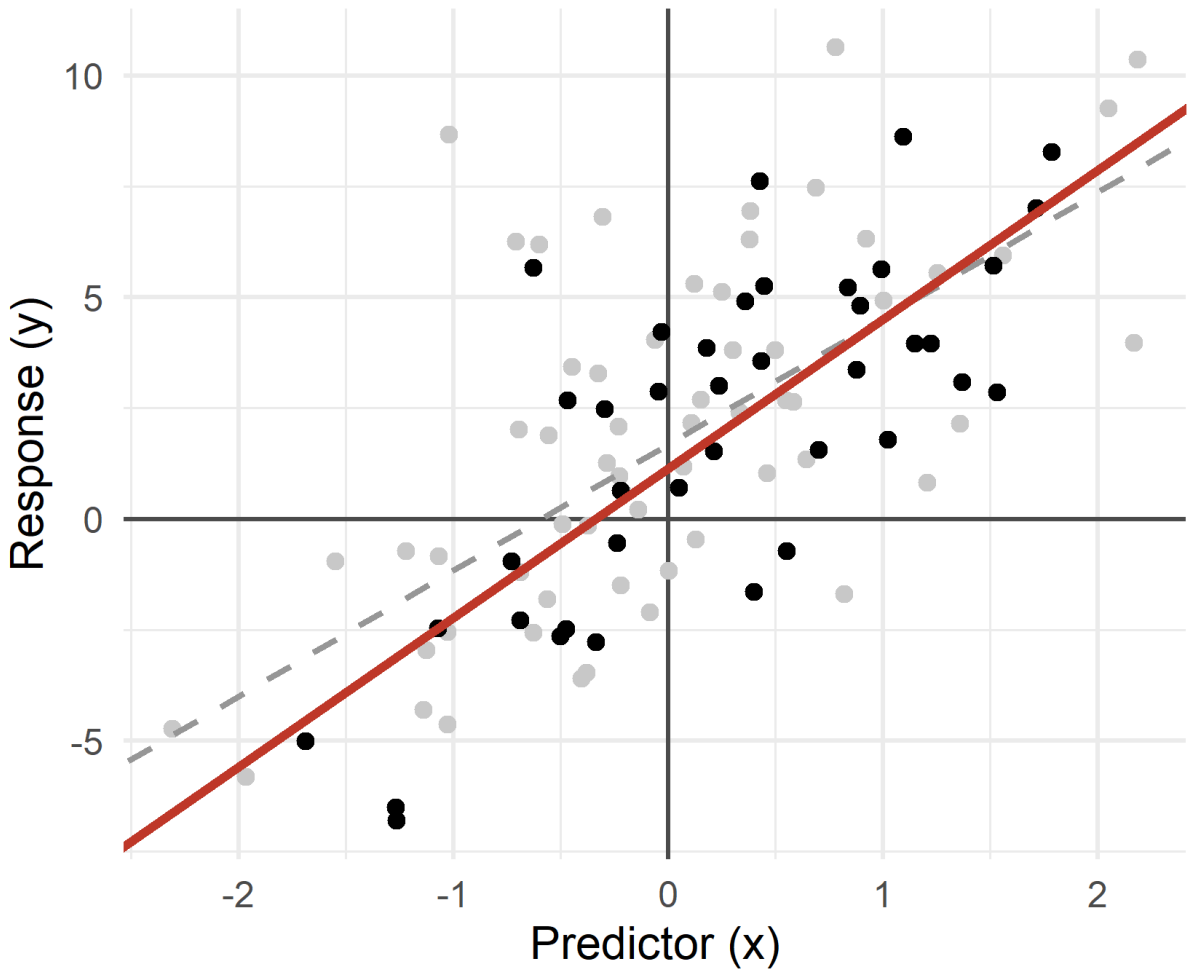
$$y_i = 1.69 + 2.84\, x_i + u_i$$

Sample 3 estimate

$$\hat{y}_i = 1.70 + 2.82\, x_i$$

Uncertainty in the estimate

A random sample of 40 from the same population.



Population regression

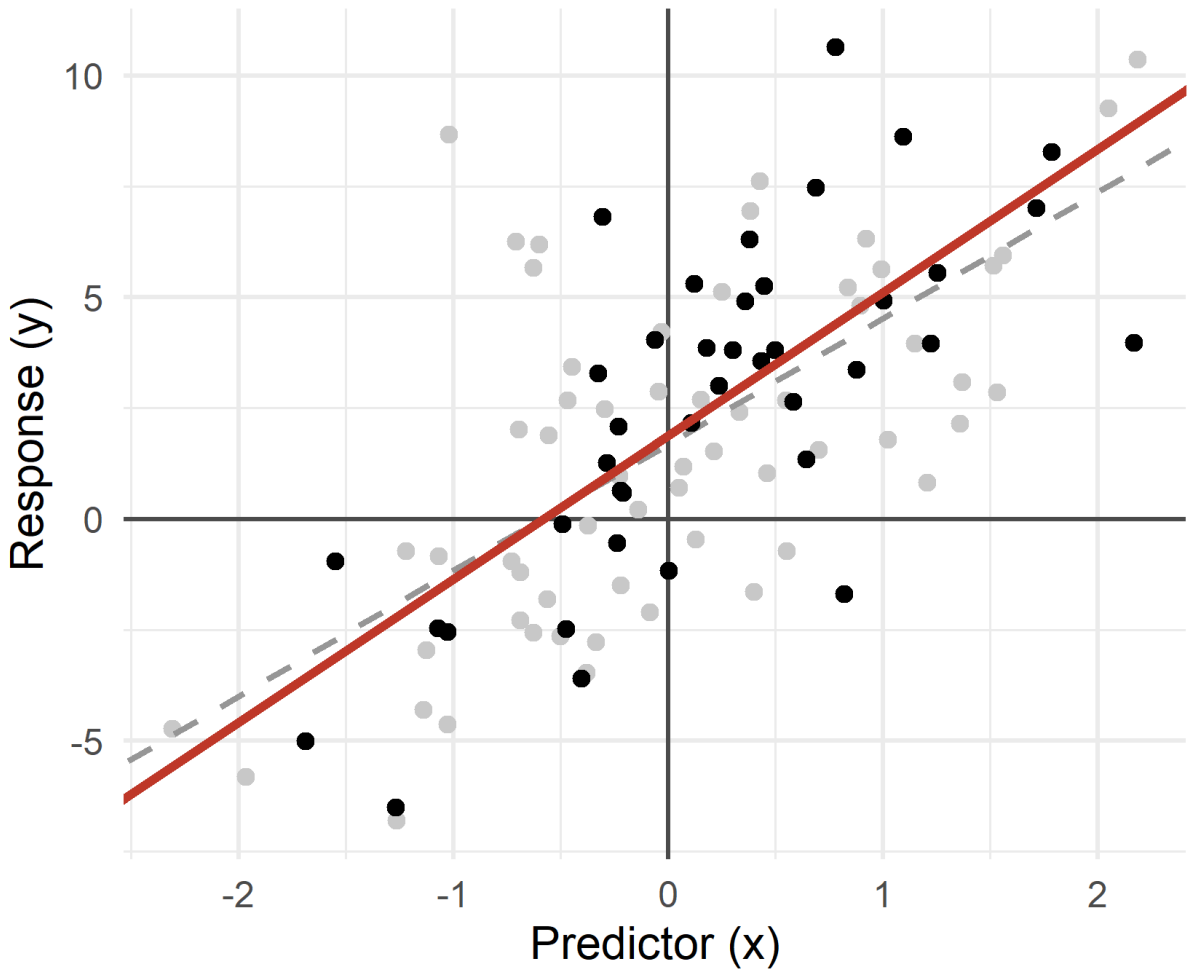
$$y_i = 1.69 + 2.84\, x_i + u_i$$

Sample 4 estimate

$$\hat{y}_i = 1.15 + 3.36\, x_i$$

Uncertainty in the estimate

A random sample of 40 from the same population.



Population regression

$$y_i = 1.69 + 2.84\, x_i + u_i$$

Sample 5 estimate

$$\hat{y}_i = 1.89 + 3.23\, x_i$$

Five samples, five lines

- Every sample gives a different line. None of them is the population line exactly.
- The estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ depend on which 40 points we happened to draw.
- How far do they wander, and are they right on average? The sampler two slides on draws as many samples as you like.

Predict: a thousand samples

Now imagine drawing not five but 1,000 samples of 40 from a population whose true slope is 3, and fitting a line to each one.

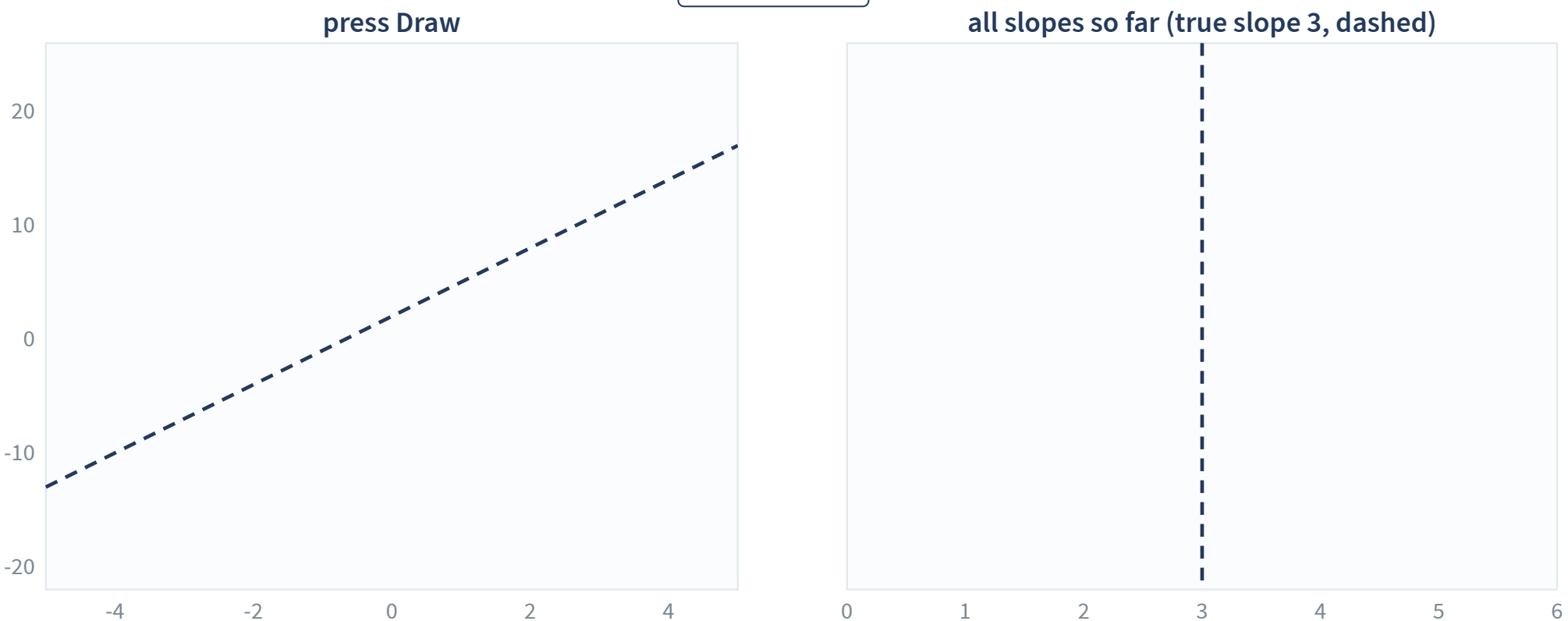
1. Where will the 1,000 slopes be centred? Compare with the true slope, 3.
2. How far will a typical slope be from that centre?
3. What shape will the histogram of the 1,000 slopes have?

Draw samples yourself

Each click draws a fresh sample from a population whose true slope is 3. Left: the sample and its line. Right: every slope drawn so far. Press Draw 100 ten times, then show the numbers. Then: what makes the slope steadier →

Sample size n 40 Noise σ 3.0 Spread of x 1.00 Draw 1 sample Draw 100

Show the numbers



samples drawn 0 mean of the slopes ? SD of the slopes ?

What the sampler showed

- Centre: the slopes pile up around 3, the true slope. On average OLS is right.
- Spread: with $n = 40$ and noise 3, a typical slope misses the true slope by about 0.5.
- Shape: a bell, symmetric around 3.
- The same picture for the population of 100 points.

One sample gives one slope. Which slope you get depends on which points you happened to draw.

The slope is a random variable

- $\hat{\beta}_0$ and $\hat{\beta}_1$ are estimators: rules that turn a sample into a number.
- So they are random variables. Their values depend on which sample we happened to draw.
- That raises the two questions every estimator must answer:
  - Is it right on average? That is, is it unbiased?
  - How much does it move from sample to sample? That is, how big is its variance?

The sampler's histogram is the sampling distribution of $\hat{\beta}_1$. Its centre and its spread are the next two sections.

2 · How much does the slope move?

Centre first, then spread. The spread has a name: the standard error.

Right on average

Under the four assumptions from last week:

1. linear in the parameters, with an additive error;
2. $E(u_i) = 0$;
3. $\mathrm{Var}(u_i) = \sigma^2$;
4. $\mathrm{Cov}(u_i, u_j) = 0$ for $i \neq j$;

OLS is unbiased:

$$E(\hat{\beta}_1) = E\left(\frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}\right) = \beta_1 \qquad \text{and} \qquad E(\hat{\beta}_0) = \beta_0$$

Assumption 2 alone is enough for unbiasedness. Assumptions 3 and 4 matter for the spread.

When is the slope biased?

A firm regresses employees' salary on their years of education, to learn what one more year of schooling is worth.

- What kinds of things end up in the error u_i?
- Are they related to education?
- Is the slope unbiased?

Ability, motivation, family connections and the quality of the school all move salary and all sit in u_i.

People with more of them also tend to study longer. Then u_i is higher, on average, for people with more education: $E(u_i)$ is not zero at every level of x.

The slope is then biased upward. It mixes the effect of schooling with the effect of everything that comes with schooling.

Three things that could make the slope steadier

On the sampler, a typical slope missed the true slope by about 0.5. Which of these would make that number smaller?

- more points in each sample?
- less scatter around the line?
- more spread in x?

Go back to the sampler and move one slider at a time.

What the sliders showed

- Raise n: the histogram gets narrower. More points pin the line down.
- Raise the noise σ: the histogram gets wider. More scatter lets many slopes fit.
- Raise the spread of x: the histogram gets narrower. Points far apart in x fix the tilt of the line, like a long plank is easier to level than a short one.
- In every setting the histogram stays centred on 3. The spread changes; the centre does not.

The variance of the slope

All three levers sit in one formula:

$$\mathrm{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum (x_i - \bar{x})^2} = \frac{\sigma^2}{S_{xx}}$$

And for the intercept:

$$\mathrm{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)$$

See the algebra for the intercept.

The standard error is the standard deviation of the estimator: $SE(\hat{\beta}_1) = \sqrt{\mathrm{Var}(\hat{\beta}_1)}$. See the algebra for the slope.

One unknown left: σ^2

We never see the errors u_i, so we do not know σ^2. We estimate it from the residuals:

$$\hat{\sigma}^2 = s^2 = \frac{\sum \hat{u}_i^2}{n-2}$$

Dividing by $n-2$ makes it unbiased: $E(s^2) = \sigma^2$.

Put it into the variance and take the square root:

$$SE(\hat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}}$$

Exercise 1 — Build the standard error

For the regression of `bikes$trips` on `bikes$temp` ($n = 781$), build the standard errors from the formula sheet.

1. Get SSE from `anova(reg)`. Compute s^2 and s.
2. Compute S_{xx} from `bikes$temp`.
3. Compute $SE(\hat{\beta}_1)$ and $SE(\hat{\beta}_0)$.
4. Check both against `vcov(reg)`.

Exercise | Start Over | Show Hint | Run Code

```
1 reg <- lm(trips ~ temp, data = bikes)
2 n <- 781
```

What the standard error says

The slope is 723.6 trips per degree, and a typical sample would miss the true slope by about 83 trips per degree.

Fit the same regression on one year at a time and the slopes are 549, 804 and 857. Each year has about a third of the days, so each yearly slope has a standard error about $\sqrt{3} \approx 1.7$ times larger: 120 to 150 trips per degree. A spread of a few hundred between years is what that predicts. See the three years.

Optional: what rescaling does to the standard error

3 · Best among its kind

OLS is unbiased. Is there an unbiased estimator that moves less?

No need to search

There are many ways to draw a line through the data, and many of them are unbiased. They differ in how much they move from one sample to the next.

We do not have to try them one by one. The answer is already known: under four simple assumptions, the unbiased estimator with the smallest variance is OLS, the one we have been using all along.

The assumptions, again

1. Linear in the parameters, with an additive error: $y_i = \beta_0 + \beta_1 x_i + u_i$.
2. The error has mean zero: $E(u_i) = 0$. On average the line is not too high or too low.
3. Constant variance: $\mathrm{Var}(u_i) = \sigma^2$. The scatter around the line is the same width everywhere (homoskedasticity).
4. Errors are uncorrelated across observations: one day's error tells you nothing about the next day's.

Assumptions 1 and 2 give unbiasedness. Adding 3 and 4 gives the next result.

The Gauss–Markov theorem

Under assumptions 1–4, among all linear and unbiased estimators, OLS has the smallest variance:

$$\mathrm{Var}(\hat{\beta}_1) \leq \mathrm{Var}(\tilde{\beta}_1) \qquad \text{and} \qquad \mathrm{Var}(\hat{\beta}_0) \leq \mathrm{Var}(\tilde{\beta}_0)$$

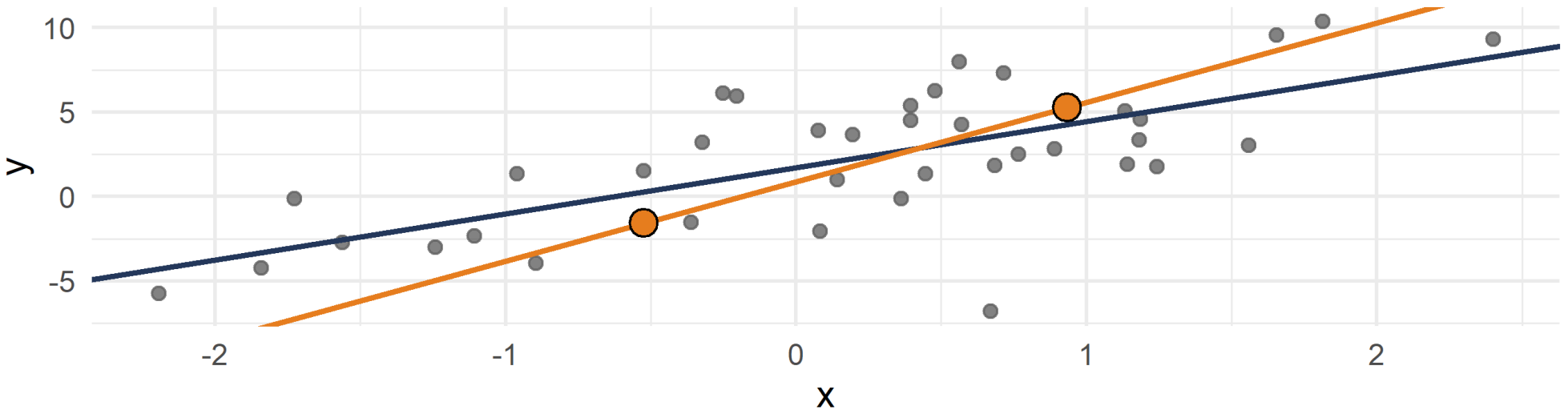
where $\tilde{\beta}_1$ and $\tilde{\beta}_0$ are any other linear and unbiased estimators.

OLS is BLUE: the Best Linear Unbiased Estimator. What “linear” means · An update from 2022

Another unbiased slope: use only two points

The simplest idea: take the first two observations and draw the line through them. Ignore all the others.

$$\tilde{\beta}_1 = \frac{y_2 - y_1}{x_2 - x_1}$$



Orange: the line through points 1 and 2, slope 4.70. Blue: OLS on all 40 points, slope 2.74.

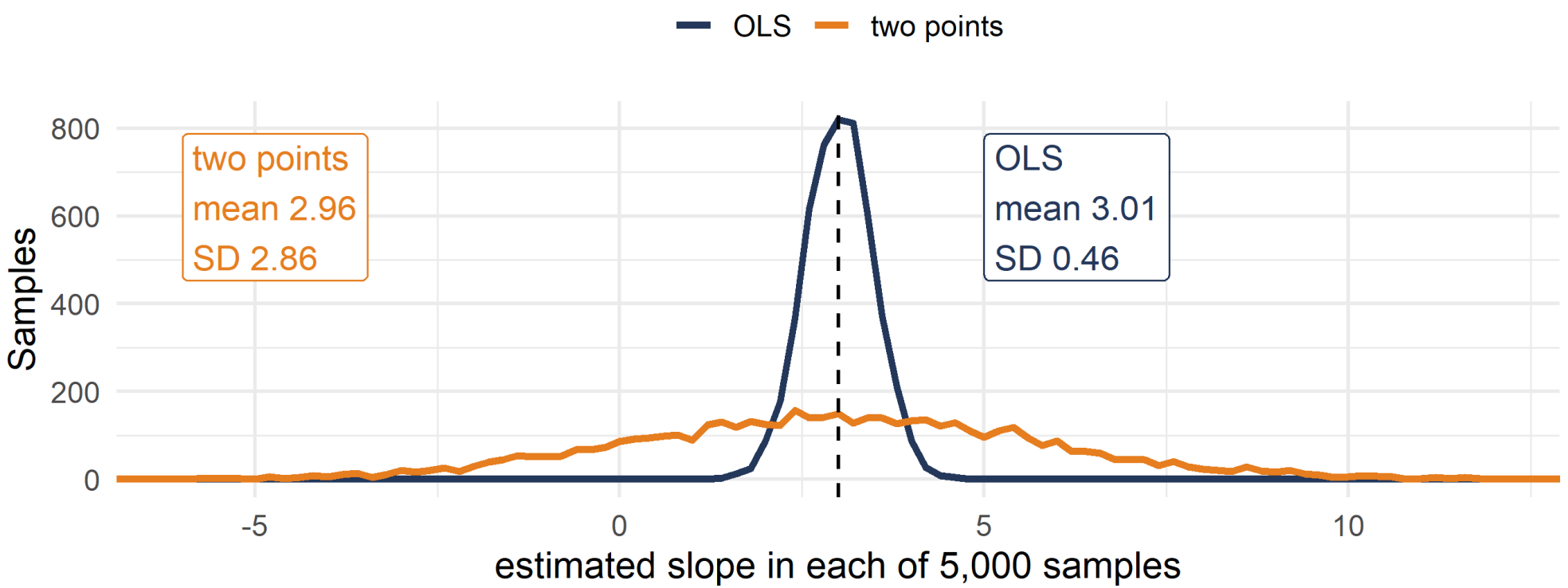
Is it unbiased? Each point has $E(y_i) = \beta_0 + \beta_1 x_i$, so

$$E(\tilde{\beta}_1) = \frac{(\beta_0 + \beta_1 x_2) - (\beta_0 + \beta_1 x_1)}{x_2 - x_1} = \frac{\beta_1 (x_2 - x_1)}{x_2 - x_1} = \beta_1$$

Yes. On average it lands on the true slope, just like OLS.

Which one moves more?

Both estimators are unbiased. Keep the same 40 x values, draw new errors 5,000 times (true slope 3), and compute both slopes in every sample. Are both centred on 3? Which curve is wider?



Both average 3: OLS 3.01, two points 2.96. But the standard deviation is 0.46 for OLS and 2.86 for the two-point line, about 6 times as large. The two-point line throws away 38 of the 40 points; OLS uses all of them.

Unbiased is not enough. Among linear unbiased estimators, OLS has the smallest variance. The theorem in a picture.

4 · The shape of the slope’s distribution

Centre and spread are settled. To test and to build intervals we also need the shape.

What we have not needed so far

We have not assumed anything about the distribution of the data or of $\hat{\beta}$.

- We did not need it to compute $\hat{\beta}_0$ and $\hat{\beta}_1$.
- We did not need it to predict $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$.
- We did not need it for the mean and the variance of the coefficients.
- We did not need it for Gauss–Markov.

For inference — confidence intervals and tests — we need the distribution of $\hat{\beta}$.

The normality assumption

We add one assumption: the errors are normal, $u_i \sim N(0, \sigma^2)$.

- Neither y_i nor x_i needs to be normal.
- But with normal errors, at each value of x the outcome is normal: $y_i \mid x_i \sim N(\beta_0 + \beta_1 x_i,\ \sigma^2)$.
- For some results this can be relaxed in large samples, thanks to the CLT.

The distributions

A linear combination of normal variables is normal. $\hat{\beta}_1$ is a weighted sum of the y_i, so:

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{\sum(x_i - \bar{x})^2}\right) \qquad \hat{\beta}_0 \sim N\left(\beta_0, \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{\sum(x_i - \bar{x})^2}\right)\right)$$

And the variance estimate has its own distribution:

$$\frac{(n-2)\,s^2}{\sigma^2} \sim \chi^2_{n-2}$$

These are the lines on your formula sheet under Simple Linear Regression.

Big samples: the CLT

With many observations we can drop the normality assumption. By the Central Limit Theorem, approximately:

$$\hat{\beta}_1 \xrightarrow{d} N\left(\beta_1,\ \frac{\sigma^2}{S_{xx}}\right) \qquad \hat{\beta}_0 \xrightarrow{d} N\left(\beta_0,\ \sigma^2\left(\frac{1}{n}+\frac{\bar{x}^2}{S_{xx}}\right)\right)$$

Your histogram of 1,000 slopes was this bell. Why a sum of errors turns normal.

5 · Testing a slope

Is the effect real, or could it be zero? Is it the number someone promised?

Why test in regression? Advertising

A retailer wants to know if advertising raises sales:

$$\text{Sales}_i = \beta_0 + \beta_1 \cdot \text{AdSpend}_i + u_i$$

Does more advertising actually raise sales? If not, the budget goes elsewhere.

$$H_0 : \beta_1 \leq 0 \text{ (ads do not raise sales)} \qquad \text{against} \qquad H_A : \beta_1 > 0 \text{ (ads raise sales)}$$

Why test in regression? Supply chain

A logistics firm models delivery time on distance to the warehouse:

$$\text{ShipDays}_i = \beta_0 + \beta_1 \cdot \text{Distance}_i + u_i$$

where distance is in kilometres and ShipDays is the average days to delivery.

Its delivery promises assume each extra 100 kilometres adds exactly one day, so one extra kilometre adds 1/100 of a day. Is that right?

$$H_0 : \beta_1 = 1/100 \qquad \text{against} \qquad H_A : \beta_1 \neq 1/100$$

If distance matters more, promises are broken. If it matters less, the firm quotes delivery times that are too slow and loses customers.

Why test in regression? A recommendation system

A video app earns money by selling ads. To set its prices, it tells advertisers that a typical user watches 10 hours a week. The app now tests a new recommendation system on some of its users:

$$\text{HoursWatched}_i = \beta_0 + \beta_1 \cdot \text{NewAlgorithm}_i + u_i$$

where `HoursWatched` is weekly viewing in hours, and `NewAlgorithm` is 1 if the user saw the new system, 0 otherwise.

Question 1. Is the mean weekly viewing of users without the new system really 10 hours, the number the app gives advertisers? If it is lower, the app overcharges them. If it is higher, the app undercharges them.

$$H_0 : \beta_0 = 10 \qquad \text{against} \qquad H_A : \beta_0 \neq 10$$

Question 2. Does mean viewing differ between users with and without the new system? This is a test for a difference in means, done by regression.

$$H_0 : \beta_1 = 0 \qquad \text{against} \qquad H_A : \beta_1 \neq 0$$

The test statistic

To test $H_0 : \beta_1 = \beta_1^0$ against $H_A : \beta_1 \neq \beta_1^0$, where β_1^0 is the value under the null:

$$T = \frac{\hat{\beta}_1 - \beta_1^0}{SE(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - \beta_1^0}{s/\sqrt{S_{xx}}} \sim t_{n-2}$$

For the intercept, $H_0 : \beta_0 = \beta_0^0$ against $H_A : \beta_0 \neq \beta_0^0$:

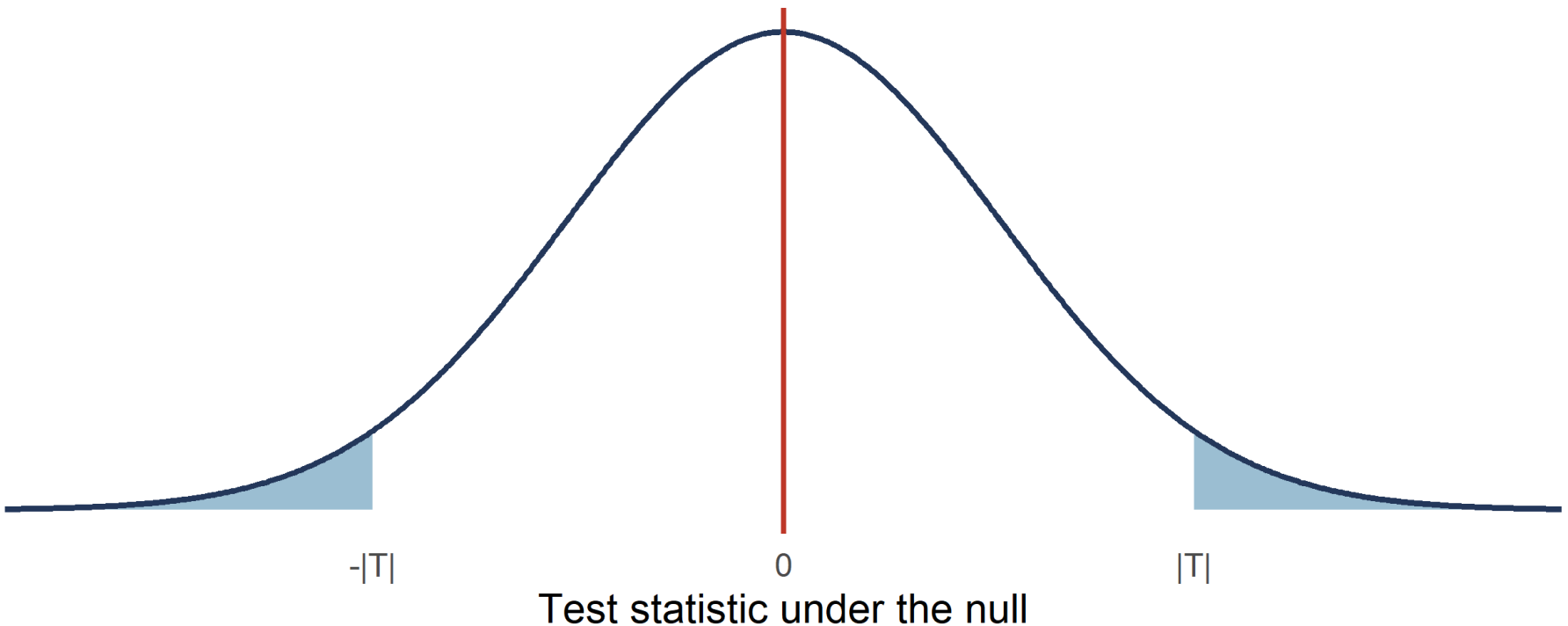
$$T = \frac{\hat{\beta}_0 - \beta_0^0}{SE(\hat{\beta}_0)} = \frac{\hat{\beta}_0 - \beta_0^0}{s\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}} \sim t_{n-2}$$

From here it is the testing procedure from W06: a critical value from `qt()`, or a p-value from `pt()`.

The p-value

The p-value is the probability of a test statistic at least as extreme as ours, if the null hypothesis were true. Regression tests are usually two-sided:

$$p\text{-value} = 2 \cdot P\left(t_{n-2} > |T|\right)$$



The shaded area is the p-value. A small p-value says: if the null were true, a result like ours would be rare.

Is the temperature slope real?

Back to the finance director's second question. Could the true effect of temperature on trips be zero?

$$H_0 : \beta_1 = 0 \qquad \text{against} \qquad H_A : \beta_1 \neq 0$$

If we cannot reject H_0, the client has no evidence that warmer days bring more trips, and no reason to size the fleet by temperature at all.

Exercise 2 — Test the temperature slope

Test $H_0 : \beta_1 = 0$ against $H_A : \beta_1 \neq 0$ at the 5% level, for the regression of `bikes$trips` on `bikes$temp` ($n = 781$).

1. Compute $\hat{\beta}_1$ and its standard error.
2. Compute the t statistic.
3. Find the 5% two-sided critical value. Do you reject?
4. Compute the p-value.

Exercise | Start Over | Show Hint | Run Code

```
1 reg <- lm(trips ~ temp, data = bikes)
2 n <- 781
```

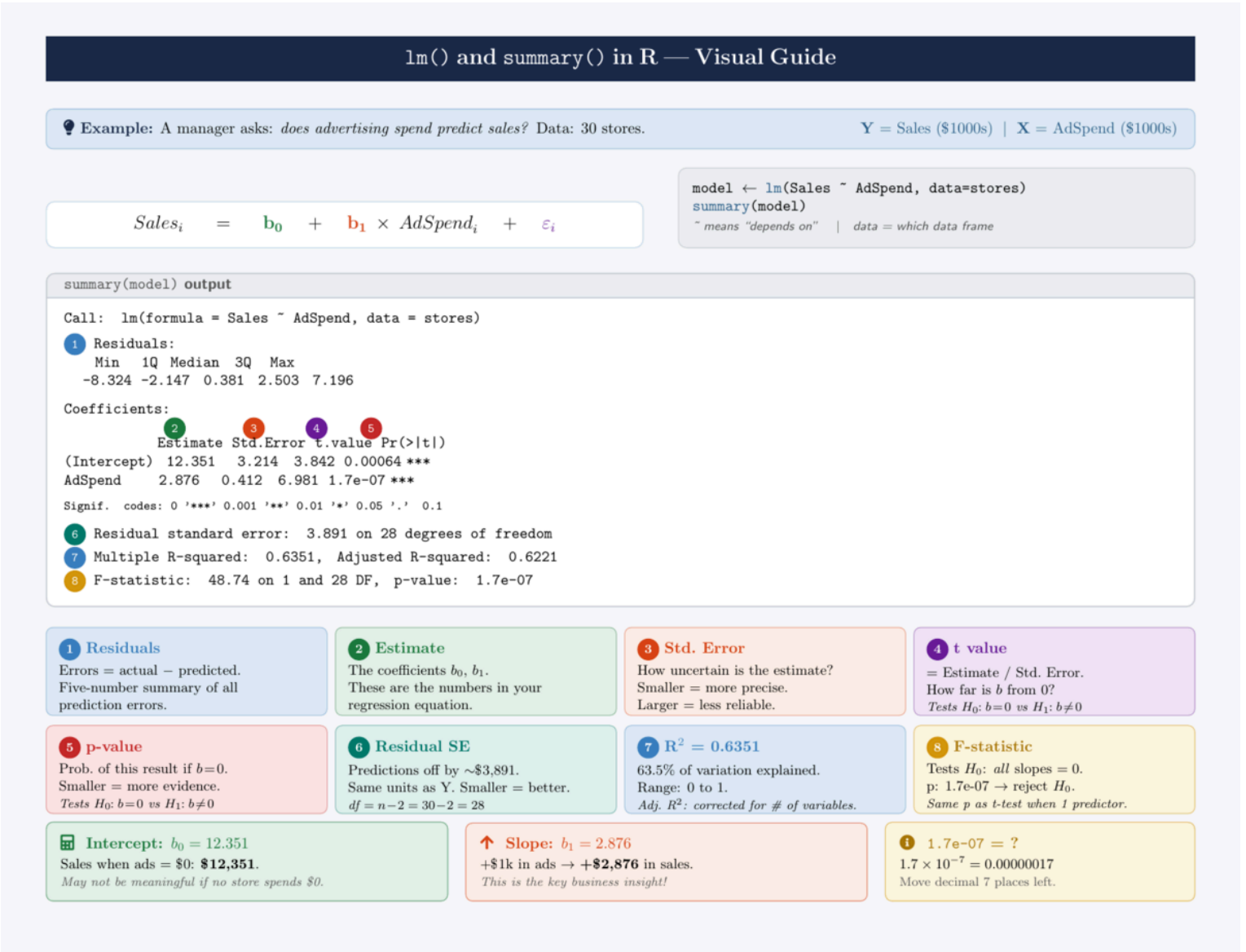
Exercise 3 — Now read it off the output

Run `summary()` on the model. Find the three numbers you just built: the standard error, the t statistic and the p-value.

Exercise | Start Over | Show Hint | Run Code

```
1 reg <- lm(trips ~ temp, data = bikes)
```

Every column, named



Does pollution cut ridership?

On days with bad air, Mexico City can declare an environmental contingency. If pollution cuts ridership, Ecobici can send fewer rebalancing trucks out on those days.

The slope of `trips` on `pm25` answers it: how many trips are lost per extra μg/m^3 of fine particles?

Exercise 4 — Pollution and trips

Regress `bikes$trips` on `bikes$pm25` ($n = 781$).

1. What is the slope? Say what it means in trips.
2. Test $H_0 : \beta_1 = 0$ against $H_A : \beta_1 \neq 0$ at the 10% level. Build T and the p-value first, then check them in `summary()`.
3. Can Ecobici plan on fewer trips on high-pollution days?

Exercise | Start Over | Show Hint | Run Code

```
1 n <- 781
```

Exercise 5 — Fridays and the sponsor

Last week: Fridays have about 1,200 fewer trips than Monday–Thursday. Now regress `bikes$trips` on `bikes$friday` ($n = 781$) and test two claims.

1. Ecobici wants to pull bikes for repair on Fridays, but only if Fridays really are quieter. Test $H_0 : \beta_1 = 0$ against $H_A : \beta_1 \neq 0$. Do you reject at 5%? At 1%?
2. The sponsor's contract assumes 30,000 trips on an average Monday–Thursday. If the true average is lower, Ecobici must refund part of the fee. The intercept is the Monday–Thursday mean. Test $H_0 : \beta_0 \geq 30{,}000$ against $H_A : \beta_0 < 30{,}000$ at 5%.

How to test the intercept, and how to get a one-sided p-value: see the appendix.

Exercise | Start Over | Show Hint | Run Code

```
1 n <- 781
```

Exercise 6 — The planning guide's 1,000

The bike maker's planning guide says each extra degree brings 1,000 more trips. Your client will order bikes by this guide unless the Mexico City data disagree.

Test $H_0 : \beta_1 = 1{,}000$ against $H_A : \beta_1 \neq 1{,}000$ at 5%, for `bikes$trips` on `bikes$temp` ($n = 781$). Should she order by the guide?

Why two-sided here: see the appendix. One more test like this one: the agent's 200 pesos.

Exercise | Start Over | Show Hint | Run Code

```
reg <- lm(trips ~ temp, data = bikes)
n <- 781
```

Exercise 7 — “More than 2,500 a year”

The tech company from last week wants to put a line in its job ads: “Here, each year of experience is worth more than 2,500 pesos a month.” Legal will approve it only if the data support it.

`hires` has 120 engineers: `hires$experience` (years) and `hires$salary` (pesos a month).

1. Test $H_0 : \beta_1 \leq 2{,}500$ against $H_A : \beta_1 > 2{,}500$ at 5%.
2. Now test the two-sided $H_0 : \beta_1 = 2{,}500$ against $H_A : \beta_1 \neq 2{,}500$ at 5%. Does the decision change?
3. Which test fits Legal’s question, and can the ad run?

Exercise | Start Over | Show Hint | Run Code

```
1 n <- 120
```

6 · A range for the slope

A test says yes or no. An interval says how big.

The confidence interval for a coefficient

T has a t_{n-2} distribution, so

$$P\left(-t_{n-2,\,1-\alpha/2} < \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} < t_{n-2,\,1-\alpha/2}\right) = 1-\alpha$$

Rearranged for β_1:

$$CI_{\beta_1} = \hat{\beta}_1 \pm t_{n-2,\,1-\alpha/2} \cdot \frac{s}{\sqrt{S_{xx}}}$$

And for the intercept:

$$CI_{\beta_0} = \hat{\beta}_0 \pm t_{n-2,\,1-\alpha/2} \cdot s\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}$$

Exercise 8 — How precisely do we know 724?

Build the 95% confidence interval for the slope of `bikes$trips` on `bikes$temp` ($n = 781$).

1. What are the lower and upper limits?
2. Does the interval contain 0? Does it contain 1,000?
3. Your client plans for the worst case. How many extra trips per degree is that?

Exercise | Start Over | Show Hint | Run Code

```
1 reg <- lm(trips ~ temp, data = bikes)
2 n <- 781
```

What the interval says

If we drew many samples and built this interval from each, 95% of the intervals would contain the true slope. So we are 95% confident that the true slope lies between 560 and 887 trips per degree.

The interval and the test agree. 0 is outside the interval: we rejected $\beta_1 = 0$. 1,000 is outside: we rejected $\beta_1 = 1{,}000$. Any value inside the 95% interval would survive a two-sided test at 5%.

One interval answers every two-sided question at once. That is why reports print intervals.

Exercise 9 — The pollution interval

For the regression of `bikes$trips` on `bikes$pm25` ($n = 781$):

1. Build the 90% confidence interval for the slope.
2. Does it agree with your test at 10% in Exercise 4?
3. In the worst case for ridership, how many trips does one extra μg/m^3 cost?

Exercise | Start Over | Show Hint | Run Code

```
1 n <- 781
```

7 · Predicting the average day

From the slope to a forecast. First for the average of many days.

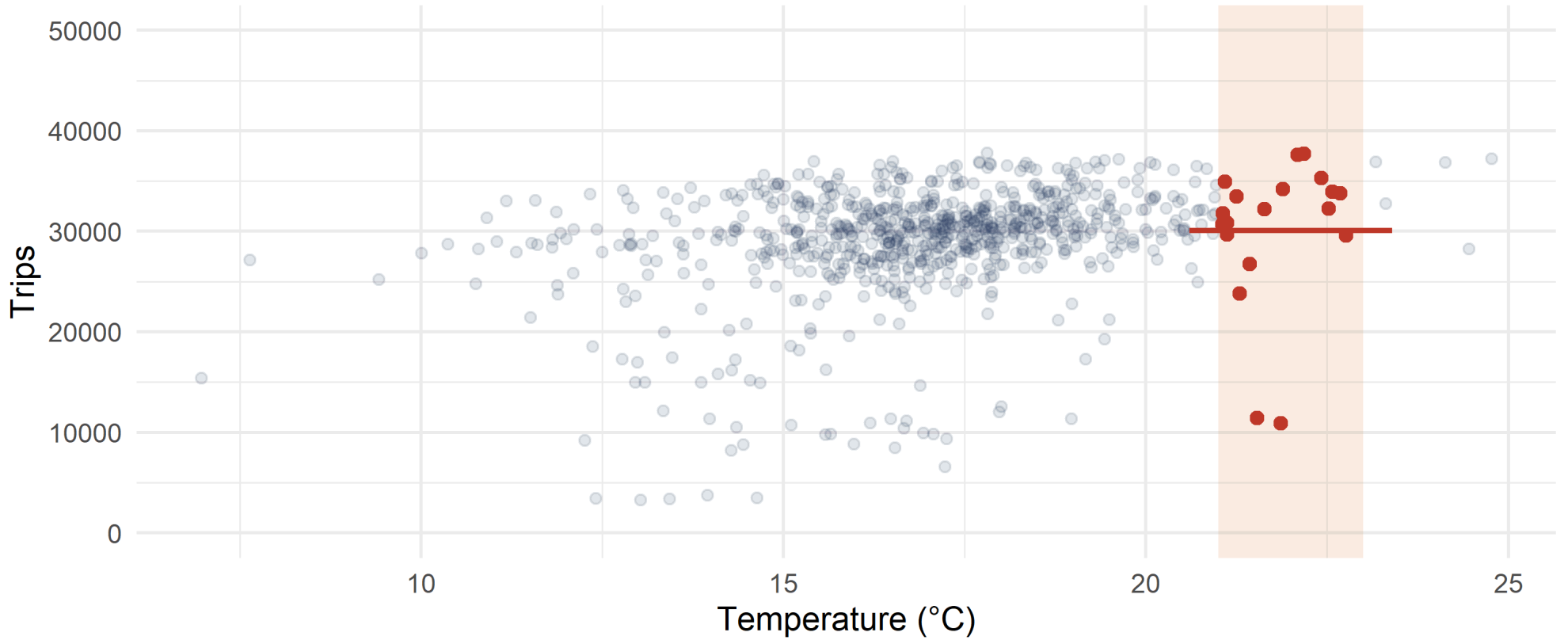
Two different questions about 22 °C

Your client runs Ecobici. Tomorrow will be 22 °C, and so will many days this season. She has two decisions to make:

1. The season budget. She pays the rebalancing crews for the whole season, so she needs the average number of trips on a 22 °C day.
2. Tonight. She has to decide how many bikes to put in the docks for tomorrow, so she needs the number of trips on that one day.

Which of the two numbers can you give more precisely: the average of many 22 °C days, or the trips on one 22 °C day? Look at the days in the data that were close to 22 °C.

What the 22 °C days show



One day near 22 °C had 10,941 trips; another had 37,721. Tomorrow could land anywhere in that range: the second question is hard.

Their average is pinned down far better: its standard error is $7{,}459/\sqrt{19} = 1{,}711$ trips. The regression uses all 781 days, so its average at 22 °C is tighter still. Same point forecast, $\hat{\beta}_0 + \hat{\beta}_1 \times 22$; different uncertainty. Compute these numbers yourself

Why the average is easier

Every day is the line plus that day's own error:

$$\text{trips}_i = \beta_0 + \beta_1 \cdot \text{temp}_i + u_i$$

- u_i is everything else that moves trips that day: a match at the stadium, a metro strike, an afternoon of rain.
- Over many days at 22 °C these errors go up and down and cancel out, since $E(u_i) = 0$. The average of many 22 °C days is just the line at 22 °C: $\beta_0 + \beta_1 \times 22$.
- So for the season budget, the only uncertainty left is how well we estimated the βs.

One day keeps its error. That is section 8: mean response or new response?

The mean response

The average number of trips at $x = x_0$ is estimated by

$$\hat{y}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0$$

Its expectation is the true average:

$$E\left(\hat{y}(x_0)\right) = \beta_0 + \beta_1 x_0$$

Its variance:

$$\operatorname{Var}\left(\hat{y}(x_0)\right) = \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)$$

Why the daily error u is not in this variance, and why it stays for one day: see the derivation.

With normal errors: $\hat{y}(x_0) \sim N\left(\beta_0 + \beta_1 x_0,\ \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)\right)$ — the Average (mean) prediction line on your formula sheet.

The confidence interval for the mean response

As before: estimate $\pm$ critical value $\times$ standard error.

$$CI_{\hat{y}(x_0)} = \hat{\beta}_0 + \hat{\beta}_1 x_0 \pm t_{n-2,\,1-\alpha/2} \cdot s\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

Exercise 10 — The average 22 °C day

For `bikes$trips` on `bikes$temp` ($n = 781$), build the 95% confidence interval for the mean number of trips on days at 22 °C.

1. Compute the point forecast.
2. Compute the standard error of the mean response. You need s, $\bar{x}$ and S_{xx}.
3. Build the interval. Then check it with `predict()`.

How to use `predict()`: give it the fitted model and a small data frame with the value of x to predict at. The column must have the same name as in the model, here `temp`.

```
new_data <- data.frame(temp = 22)

predict(reg, new_data, interval = "confidence", level = 0.95)
```

It prints three numbers: `fit` is the point forecast, and `lwr` and `upr` are the two ends of the interval.

Exercise | Start Over | Show Hint | Run Code

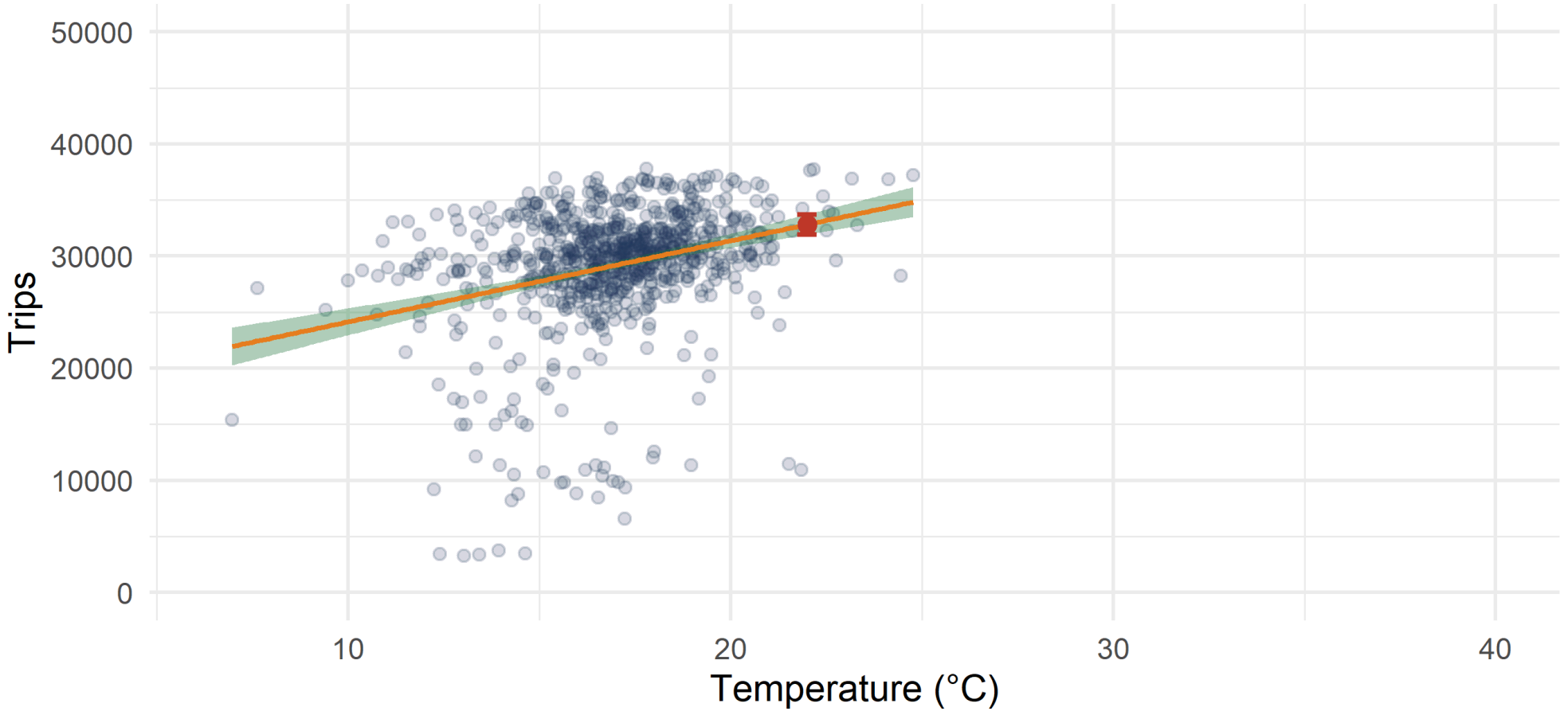
```
reg <- lm(trips ~ temp, data = bikes)
n <- 781
x0 <- 22
```

What the interval means

If we took many samples and built this interval from each, 95% of them would contain the true average number of trips on 22 °C days.

We are 95% confident that the average 22 °C weekday brings between 31,907 and 33,715 trips. That is what she budgets the season against.

The band around the line

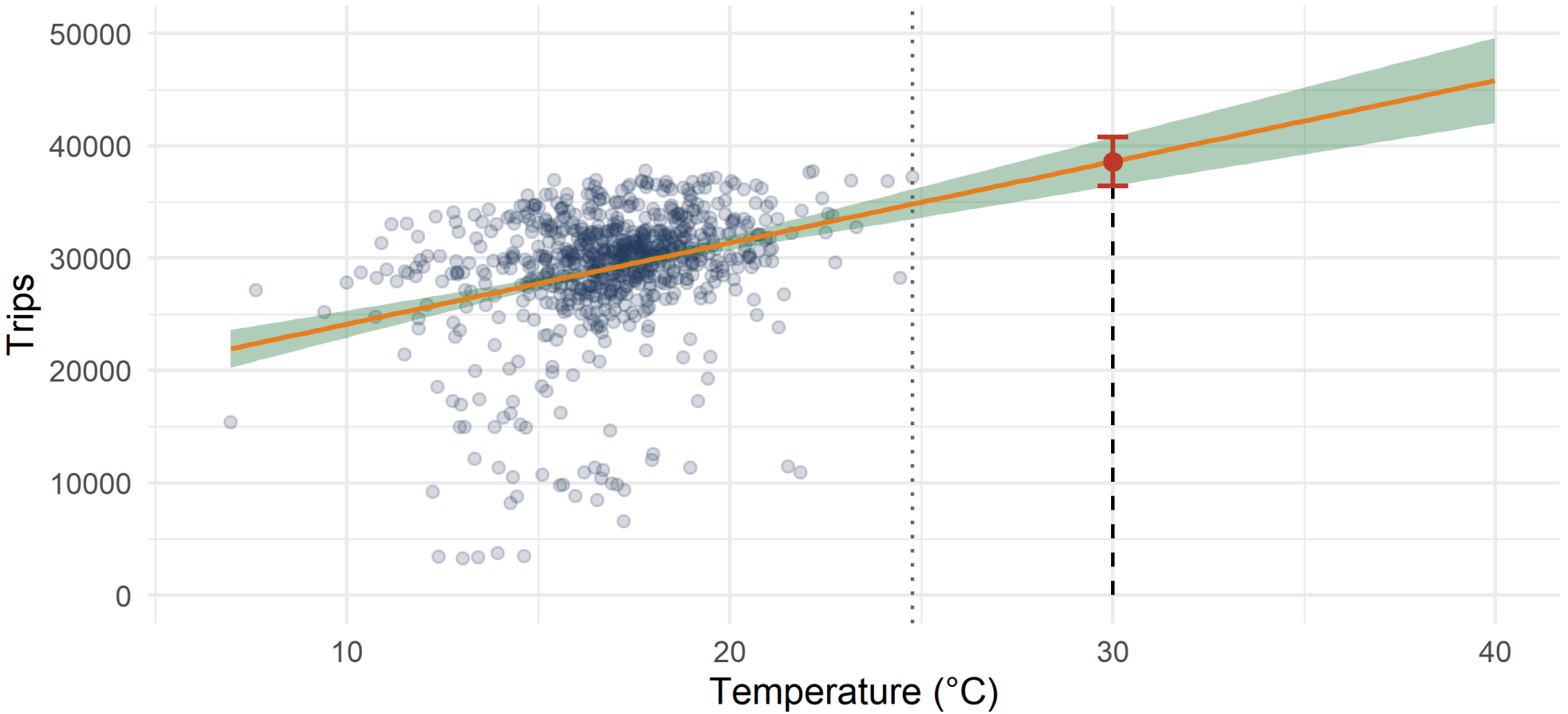


Where is the band narrowest? Why?

At $\bar{x} = 17.0$ °C, where $(x_0 - \bar{x})^2 = 0$. Away from the centre, any error in the slope tilts the line further from the truth.

A heat wave at 30 °C?

The forecast for next summer includes days at 30 °C. The hottest weekday in the data was 24.8 °C.



The 95% interval for the mean at 30 °C is 38,599 ± 2,165 trips — 2.4 times as wide as at 22 °C, and almost six times as wide as at 17 °C.

8 · Predicting one day

The average of many days is not tomorrow. One day carries its own error.

Mean response or new response?

Both of your client's questions start from the same number, $\hat{\beta}_0 + \hat{\beta}_1 \times 22$.

- Mean response, the season budget: many 22 °C days. Their errors cancel out, so only the error in the βs is left. Back to why.
- New response, tonight's docks: one day, tomorrow. Its error u_0 has nothing to cancel against.

The new response

New response. The number of trips tomorrow, one day at 22 °C. This decides how many bikes go in the docks tonight.

- With one day, its error u_0 does not cancel. Tomorrow may bring the match, the strike or the rain.
- In the mean response those surprises average out. Here they stay.
- So there is more uncertainty, and a wider interval: $\mathrm{Var}(\text{new response}) = \mathrm{Var}(\text{mean response}) + \mathrm{Var}(u_0)$.
- This interval needs the errors to be normal. The CLT is not enough, because the interval is about one error u_0, not about an average.

The prediction interval

For a new observation at x_0, the error we make is $y_0 - \hat{y}(x_0)$:

$$\operatorname{Var}\left(y_0 - \hat{y}(x_0)\right) = \operatorname{Var}\left(\hat{\beta}_0 + \hat{\beta}_1 x_0\right) + \operatorname{Var}(u_0) = \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)$$

So, the New observation line on your formula sheet: $y(x_0) \sim N\left(\beta_0 + \beta_1 x_0,\ \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)\right)$

And the interval:

$$PI_{y(x_0)} = \hat{\beta}_0 + \hat{\beta}_1 x_0 \pm t_{n-2,\,1-\alpha/2} \cdot s\sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

Exercise 11 — Tomorrow at 22 °C

Tomorrow will be 22 °C. Build the 95% prediction interval for the number of trips tomorrow, for `bikes$trips` on `bikes$temp` ($n = 781$).

1. Compute the standard error for a new observation.
2. Build the interval. Check it with `predict()`.
3. How many times wider is it than the interval for the average 22 °C day?

Exercise | Start Over | Show Hint | Run Code

```
1 reg <- lm(trips ~ temp, data = bikes)
2 n <- 781
3 x0 <- 22
```

Exercise 12 — Near the centre and far from it

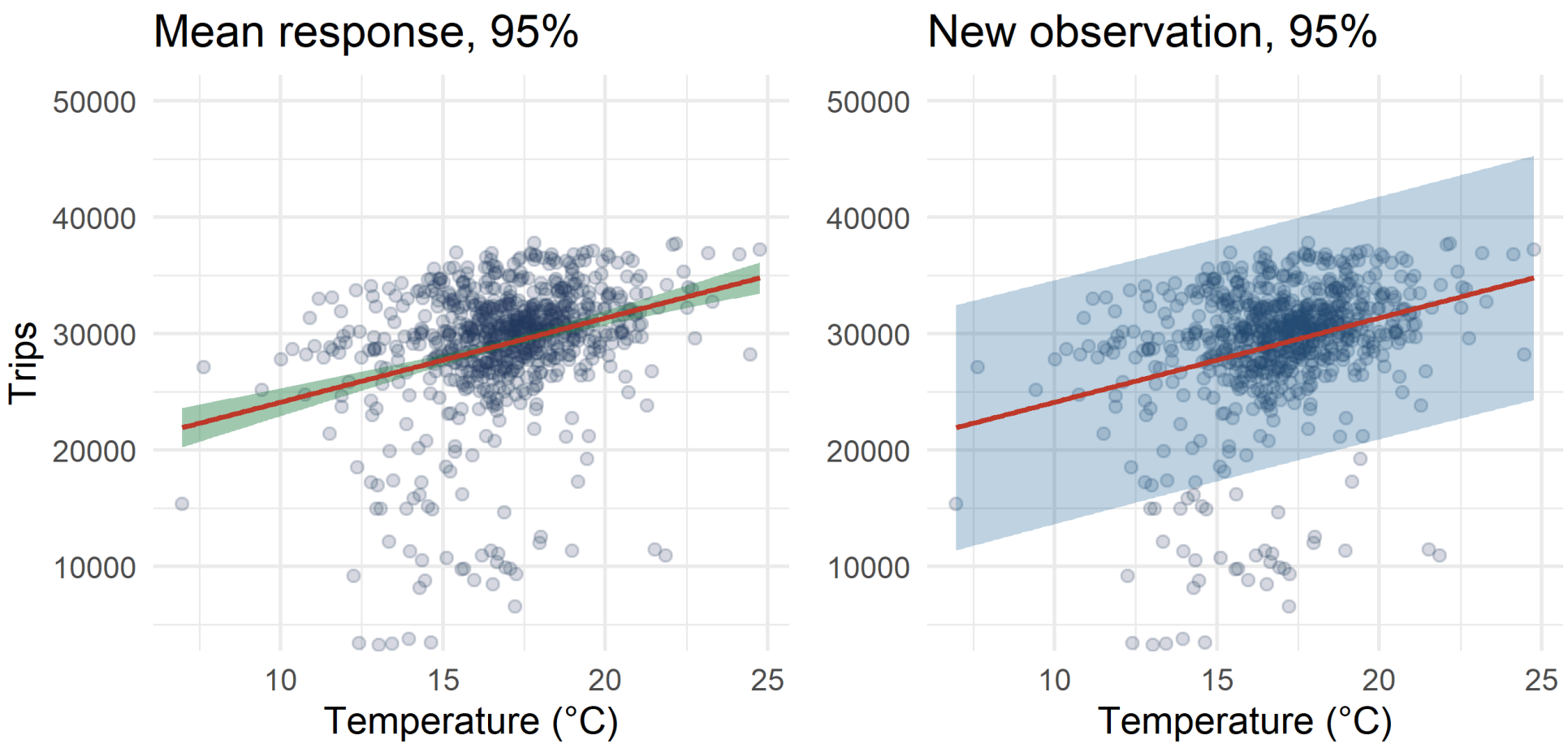
Compute both 95% intervals — for the mean and for one new day — at $x_0 = 17$ °C (the centre of the data) and at $x_0 = 22$ °C. Use `predict()`.

1. How wide is each of the four intervals?
2. Moving from 17 °C to 22 °C, which interval grows more, in relative terms?
3. Why?

Exercise | Start Over | Show Hint | Run Code

```
1 reg <- lm(trips ~ temp, data = bikes)
```

Two bands


Mean response, 95%
New observation, 95%
Trips
50000
40000
30000
20000
10000
10
15
20
25
Temperature (°C)

- Left: where the average day lies. A thin band that widens away from 17 °C.
- Right: where one day lies. A wide band, almost the same width everywhere.
- The points below the right band are holidays and strike days. The prediction interval assumes normal errors, and those days break that assumption. W09 checks it.

Exam spotlight — Spring 2026 Midterm 2

Midterm 2 questions on the fintech app data: 500 users, `fintech$MonthlyVolume` (thousands of pesos a month) on `fintech$AccountAge` (months on the platform).

→ More exam practice at the end of this deck

Exam question — Q10, Q11 & Q12

Q10. State the hypotheses for testing whether AccountAge has any effect on MonthlyVolume. Show how the t statistic is calculated. At 5%, do you reject? Use both the t statistic and the p-value.

Q11. The product manager believes each extra month raises monthly volume by more than 1.5 thousand pesos. Test $H_0 : \beta_1 = 1.5$ against $H_A : \beta_1 > 1.5$ at 5%.

Q12. An industry report claims each extra month adds exactly 2.5 thousand pesos. Test $H_0 : \beta_1 = 2.5$ against $H_A : \beta_1 \neq 2.5$ at 5%, with $t_{498,\,0.975} \approx 1.965$.

Exercise | Start Over | Show Hint | Run Code

```
1  n <- 500
```

Exam question — Q14 & Q15

Q14. Write the formula for a 95% confidence interval for β_1. Compute it using $t_{498,\,0.975} \approx 1.965$. Does 2.5 (the industry benchmark from Q12) fall inside? Is that consistent with your test?

Q15. Build a 90% confidence interval for the intercept β_0, using $t_{498,\,0.95} \approx 1.648$. Interpret it.

Exercise | Start Over | Show Hint | Run Code

```
1  n <- 500
```

Exam question — Q16 to Q19

Use $n = 500, \hat{\sigma} = 61.46, \bar{x} = 31.464, S_{xx} = 150{,}582.4$ and $t_{498,\,0.975} \approx 1.965$.

Q16. Predict monthly volume for a user with 36 months on the platform. Why would a prediction at 120 months be unreliable?

Q17. Build the 95% confidence interval for the average volume of users with 36 months.

Q18. Build the 95% prediction interval for one specific user with 36 months. Why is it wider?

Q19. Your colleague says: “The prediction interval for a new observation is always wider than the confidence interval for the mean response.” You say: “The difference in width depends only on the sample size.” (a) Both correct. (b) Your colleague is correct, you are not. (c) You are correct, your colleague is not. (d) Neither is correct.

Exercise | Start Over | Show Hint | Run Code

```
1 n <- 500
2 s <- 61.46
3 xbar <- 31.464
4 Sxx <- 150582.4
```

→ More exam practice at the end of this deck

9 · Putting it together

One business question, answered with a number, a picture and one sentence.

🖊️ Exercise 13 — Is the extra room worth it, and how sure are you?

The landlord again (`rents$size` in m^2, `rents$rent` in pesos a month, $n = 150$). Adding 10 m^2 costs 150,000 pesos.

1. A number: the 95% confidence interval for the extra monthly rent from 10 m^2. Turn it into a range of payback times, in months.
2. A picture: one plot she can read in three seconds.
3. One sentence for the landlord.

Exercise | Start Over | Show Hint | Run Code

```
1
```

Summary: simple linear regression

Concept	Key formula or idea
Model	$y_i = \beta_0 + \beta_1 x_i + u_i$
OLS	Minimises $SSE = \sum (y_i - \hat{y}_i)^2$
Slope	$\hat{\beta}_1 = \mathrm{Cov}(x, y)/\mathrm{Var}(x)$
R^2	Share of the variation in y explained by x
Standard error	$SE(\hat{\beta}_1) = s/\sqrt{S_{xx}}$, with $s^2 = SSE/(n-2)$
Test	$T = (\hat{\beta}_1 - \beta_1^0)/SE(\hat{\beta}_1) \sim t_{n-2}$
CI for β_1	$\hat{\beta}_1 \pm t_{n-2,\,1-\alpha/2} \cdot SE(\hat{\beta}_1)$
CI for the mean at x_0	$\hat{y}(x_0) \pm t_{n-2,\,1-\alpha/2} \cdot s\sqrt{1/n + (x_0 - \bar{x})^2/S_{xx}}$
PI for one new y at x_0	$\hat{y}(x_0) \pm t_{n-2,\,1-\alpha/2} \cdot s\sqrt{1 + 1/n + (x_0 - \bar{x})^2/S_{xx}}$

Key takeaways

- OLS finds the line that minimises the squared residuals.
- $\hat{\beta}_1$ is the change in y for a one-unit change in x, and it moves from sample to sample.
- Its standard error shrinks with more data, less noise and more spread in x.
- We test and build intervals for coefficients with the t distribution, $n - 2$ degrees of freedom.
- Intervals for the mean response are narrower than for a new response, which carries its own error.

What you should be able to do

- Explain why $\hat{\beta}_1$ is a random variable, and what makes its standard error small.
- Build s, S_{xx} and $SE(\hat{\beta}_1)$ by hand, and check with `vcov()`.
- State both hypotheses for any test on β_0 or β_1, two-sided or one-sided, with a null value of 0 or any other number.
- Compute T, the critical value with `qt()` and the p-value with `pt()`; find the same numbers in `summary()`.
- Build a confidence interval for a coefficient, and read a test off it.
- Build the confidence interval for a mean response and the prediction interval for one new observation, by hand and with `predict()`, and say why one is wider.
- Say when a prediction is an extrapolation.

Next week: are the assumptions behind all of this true? Residual plots, non-constant variance, non-linearity and normality.

Go to the exam practice →

Practice for this week

Weekly exercise set: `W08_exercises.pdf`, with solutions in `W08_solutions.pdf`.

You can already solve exercises 1–21.

Suggested before the next class, closest to the exam:

Exercise	Topic	How
4	Murders and poverty: reading regression output	R as a calculator
15	CEO pay and performance: standard error and tests by hand (`ceosal1`)	R with the data
16	Fair tax assessments: tests against 1 and 0, the mean at 300 (`hprice1`)	R with the data
17	Pricing a new route: both intervals by hand (`airfare`)	R with the data
18	One more letter: a one-sided test against 2 (`charity`)	R with the data

Exam practice

Problems from this week's set. ← Back to the exam spotlight · ← Back to the summary

Exam practice 1 — CPS: the slope's interval

[P2 #4b]

Return to the `cps` regression of weekly earnings `earnwk` on education `educ`, estimated on employed workers only (`cpsemployed`, the subset where `lfstatus == "Employed"`). Provide a 95% confidence interval for the slope, and say in one sentence what it means for the return to a year of schooling.

Exercise | Start Over | Show Hint | Run Code

```
1
```

Exam practice 2 — IQ regressions

[P2 #5b]

Three regressions from the Week 7 set ($n = 930$):

	iq on momiq	iq on dadiq	iq on avgiq
$\hat{\alpha}$ (se)	69.585 (3.394)	77.156 (3.426)	56.487 (4.331)
$\hat{\beta}$ (se)	0.299 (0.034)	0.231 (0.035)	0.434 (0.044)
R^2	0.0925	0.0522	0.1161

Each IQ variable has a sample mean close to 100 and a standard deviation close to 15.

a. Build a 95% confidence interval for the slope in the model with `momiq`.

b. In the regression of `iq` on `dadiq`, test $H_0 : \beta = 0$ against $H_A : \beta \neq 0$. Argue using the p-value.

c. A mother has IQ 130 and a father IQ 85. Using the model with `avgiq`, predict their child's IQ with a 99% prediction interval.

Exercise | Start Over | Show Hint | Run Code

```
1 n <- 930
```

Exam practice 3 — Scatter and uncertainty

[P2 #8]

Is I or II higher, or are they equal? Explain. For a regression line, the uncertainty in the slope estimate $\hat{\beta}_1$ is higher when:

- i. there is a lot of scatter around the regression line, or
- ii. there is very little scatter around the regression line.

i. is higher. More scatter means a larger residual standard error s, and

$$SE(\hat{\beta}_1) = \frac{s}{\sqrt{\sum (x_i - \bar{x})^2}}$$

so the standard error of the slope grows with it: ten times the scatter gives ten times the standard error.

When points sit tightly around the line, only a narrow range of slopes fits them. When they are spread out, many different slopes fit about equally well.

Exam practice 4 — Murders and poverty

[P2 #15]

Regression of annual murders per million on the percentage living in poverty, in 20 metropolitan areas:

	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	−29.901	7.789	−3.839	0.001
poverty%	2.559	0.390	6.562	0.000

$s = 5.512, R^2 = 70.52\%, R^2_{adj} = 68.89\%.$

a. Write out the linear model.

b. Interpret the intercept.

c. Interpret the slope.

d. Interpret R^2.

e. Calculate the correlation coefficient.

f. State hypotheses for testing whether poverty percentage is a significant predictor of murder rate.

g. Report the conclusion of the test in context.

h. Construct a 95% confidence interval for the slope, and interpret it.

i. Do the hypothesis test and the confidence interval agree? Explain.

Exercise | Start Over | Show Hint | Run Code

Exam practice 5 — Cats' hearts

[P2 #16]

Regression of heart weight (g) on body weight (kg) for 144 domestic cats:

	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	−0.357	0.692	−0.515	0.607
body wt	4.034	0.250	16.119	0.000

$s = 1.452, R^2 = 64.66\%, R^2_{adj} = 64.41\%.$

a. Write out the linear model.

b. Interpret the intercept.

c. Interpret the slope.

d. Interpret R^2.

e. Calculate the correlation coefficient.

Exercise | Start Over | Show Hint | Run Code

```
1
```

Exam practice 6 — Height and weight

[P2 #17]

Weight (kg) on height (cm) for 507 physically active individuals:

	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	−105.0113	7.5394	−13.93	0.0000
height	1.0176	0.0440	23.13	0.0000

a. Describe the relationship between height and weight.

b. Write the regression equation. Interpret the slope and the intercept in context.

c. Do the data provide strong evidence that an increase in height is associated with an increase in weight? State the null and alternative hypotheses, report the p-value, and state your conclusion.

d. The correlation between height and weight is 0.72. Calculate R^2 and interpret it.

Exercise | Start Over | Show Hint | Run Code

```
1 n <- 507
```

Exam practice 7 — Beer and blood alcohol

[P2 #18]

Sixteen student volunteers at Ohio State University drank a randomly assigned number of cans of beer. Thirty minutes later their blood alcohol content (BAC) was measured.

	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	−0.0127	0.0126	−1.00	0.3320
beers	0.0180	0.0024	7.48	0.0000

a. Describe the relationship between number of beers and BAC.

b. Write the regression equation. Interpret the slope and the intercept.

c. Test whether the number of beers significantly predicts BAC: $H_0 : \beta = 0$ against $H_A : \beta \neq 0$. Report the p-value and conclude.

d. The correlation between beers and BAC is 0.89. Calculate R^2 and interpret it.

e. Would the relationship be as strong if we observed people in a bar (self-reported drinks, measured BAC)? Why or why not?

Exercise | Start Over | Show Hint | Run Code

```
1 n <- 16
```

Exam practice 8 — Husbands and wives

[P2 #19]

Heights of 170 married couples in Britain. Wife's height on husband's height (inches):

	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	43.5755	4.6842	9.30	0.0000
husband height	0.2863	0.0686	4.17	0.0000

a. Is there strong evidence that taller men marry taller women? State the hypotheses and your conclusion.

b. Write the regression equation.

c. Interpret the slope and the intercept in context.

d. Given $R^2 = 0.09$, what is the correlation of heights?

e. Predict the wife's height for a husband who is 69 inches tall. Comment on reliability.

f. Would it be wise to use this model for a husband who is 79 inches tall? Why or why not?

Exercise | Start Over | Show Hint | Run Code

```
1  n <- 170
```

Exam practice 9 — Low birth-weight babies

[P2 #21]

Twenty-five low birth-weight babies at a Harvard teaching hospital. Head circumference (cm) on gestational age (weeks):

$$\widehat{\text{head circumference}} = 3.91 + 0.78 \times \text{gestational age}$$

a. Predict head circumference for a baby with gestational age 28 weeks.

b. The standard error of the slope is 0.35 (df = 23). Test $H_0 : \beta = 0$ against $H_A : \beta \neq 0$. Does the model give strong evidence that gestational age is associated with head circumference?

Exercise | Start Over | Show Hint | Run Code

```
1
```

Exam practice 10 — House prices: CI versus PI

[P2 #22]

`houseprices`: one-family home sales in Ames, Iowa, 2006–2010. Outcome `houseprices$saleprice` (dollars); explanatory variable `houseprices$grlivarea` (above-ground living area, square feet).

a. Estimate the regression with `lm` and store it in `hp_results`. Report the equation.

b. Interpret the slope. Is the sign what you expected?

c. Build a 95% confidence interval for the slope. Is there significant evidence that living area is associated with sale price?

d. Use `predict()` for the 95% confidence interval for the mean sale price of houses with 2,000 square feet. Interpret it.

e. Now the 95% prediction interval for the sale price of a single house with 2,000 square feet. Interpret it.

f. Why is the prediction interval wider? Explain the source of the extra uncertainty.

g. Plot `saleprice` against `grlivarea`, and the fitted values. Does a linear model seem adequate?

Exercise | Start Over | Show Hint | Run Code

```
1 n <- 973
```

Exam practice 11 — Births: one-sided and two-sided

[P2 #23]

`births`: singleton births in December 2021, mothers aged 25–35 ($n = 50{,}249$). Outcome `births$bweight`, birthweight in grams.

a. Regress `bweight` on `smoke` (1 if the mother smoked during pregnancy). Store it in `birth_results`.

b. Interpret the intercept and the slope. What does the slope say about the average birthweight gap between smokers and non-smokers?

c. The standard concern is that smoking reduces birthweight. Test $H_0 : \beta_{\text{smoke}} = 0$ against $H_A : \beta_{\text{smoke}} < 0$ with the one-sided p-value, at $\alpha = 0.05$.

d. Now test $H_0 : \beta_{\text{smoke}} = 0$ against $H_A : \beta_{\text{smoke}} \neq 0$. Compare the two p-values. Why does the one-sided test have more power here?

e. Build a 99% confidence interval for β_{smoke}. Is zero inside? Is that consistent with your test?

f. Regress `bweight` on the mother's age `age`. Test $H_0 : \beta_{\text{age}} = 0$ against $H_A : \beta_{\text{age}} \neq 0$ at 5%. Do older mothers (within 25–35) have heavier or lighter babies?

Exercise | Start Over | Show Hint | Run Code

```
1 n <- 50249
```

Exam practice 12 — ENIGH: CI versus PI

[#12 · core]

`h` is a sample of 200 ENIGH households with positive income, already drawn for you (the sheet's `set.seed(11)` draw). `h$food_spending` and `h$monthly_income` are in pesos a month. Fit `m <- lm(food_spending ~ monthly_income, data = h)`.

a. Report the slope, its standard error and its t statistic.

b. State H_0 and H_A for the slope and give your decision at 5%, using $t_{198,\,0.975} = 1.972$.

c. Report the 95% confidence interval for the slope. Interpret it in pesos of food per peso of income.

d. Predict food spending for a household earning 20,000 pesos, with the 95% confidence interval for the mean at that income.

e. Report the 95% prediction interval for a single household at that income. Why is it so much wider?

f. The prediction interval runs from about −786 to 8,652 pesos. Food spending cannot be negative. What does this say about using the model for individual households?

Exercise | Start Over | Show Hint | Run Code

```
1
```

Exam practice 13 — Returns to education

[#13 · core]

`wage1` holds 526 US workers: `wage1$wage` is the hourly wage in dollars, `wage1$lwage` its log, and `wage1$educ` years of education.

a. Fit `lm(wage ~ educ, data = wage1)` and report the intercept, slope, standard error and R^2.

b. Interpret the slope in dollars per extra year of schooling.

c. The intercept is −0.905. Why is a negative predicted wage not evidence that the model is broken?

d. Give the 95% confidence interval for the slope. Is a return of exactly 50 cents an hour per year of schooling consistent with the data ($H_0 : \beta = 0.5$ against $H_A : \beta \neq 0.5$ at 5%)?

e. Fit `lm(lwage ~ educ, data = wage1)`. Report the slope and interpret it as a percentage return per year of schooling.

f. R^2 is only about 0.17. What does that mean for advising one student about their future wage?

Exercise | Start Over | Show Hint | Run Code

```
1 n <- 526
```

📝 Exam practice 14 — Do more hours mean more pay?

[#14 · stretch]

`ws` is a sample of 300 ENOE workers from 2022 with income between 0 and 100,000 pesos a month, already drawn for you (the sheet's `set.seed(5)` draw). `ws$Income` is monthly income in pesos and `ws$Hours_worked_lw` hours worked last week.

a. Fit `lm(Income ~ Hours_worked_lw, data = ws)`. Report the slope, its standard error, its t statistic and the p-value for $H_0 : \beta = 0$ against $H_A : \beta \neq 0$.

b. Interpret the slope: how many extra pesos a month does one extra weekly hour predict?

c. Give the 95% confidence interval for the slope.

d. Report R^2. It is very low. Does that make the slope meaningless? Explain the difference between a slope being precisely estimated and a model being predictive.

e. A consultant proposes paying staff purely by hours worked, citing this regression. Give two reasons from your output why that is a weak basis for the policy.

f. Name one variable missing from this regression that plausibly explains both hours and income, and say which direction it would bias the slope.

Exercise | Start Over | Show Hint | ▷ Run Code

```
1  n <- 300
```

Exam practice 15 — Is CEO pay tied to performance?

[#15 · core]

`ceosal1` holds 209 chief executives (CEOs) of large US firms in 1990. `ceosal1$salary` is the CEO's yearly salary in thousands of dollars. `ceosal1$roe` is the firm's return on equity in percent: the profit per 100 dollars of the owners' money. A shareholder group says that boards pay for performance: when the firm earns more, the CEO earns more. Fit `reg <- lm(salary ~ roe, data = ceosal1)`.

a. Compute the slope and the intercept with `cov()`, `var()` and `mean()`. How many dollars of salary go with one more percentage point of return on equity?

b. Build the standard error of the slope from the formula sheet: take SSE from `anova(reg)`, then compute s, S_{xx} and $SE(\hat{\beta}_1)$. Check it with `vcov(reg)`.

c. Test $H_0 : \beta_1 \leq 0$ against $H_A : \beta_1 > 0$ at 5%. Report the t statistic, the critical value and the p-value.

d. Now test $H_0 : \beta_1 = 0$ against $H_A : \beta_1 \neq 0$ at 5%. Does the decision change? Find your t statistic in `summary(reg)`. Which of your two p-values does `summary()` print?

e. Which of the two tests matches the shareholder group's claim? Read R^2 in `summary(reg)`: what share of the differences in CEO salaries does return on equity explain? Write one sentence for the group.

Exercise | Start Over | Show Hint | Run Code

```
1 n <- 209
```

Exam practice 16 — Are the tax assessments fair?

[#16 · stretch]

`hprice1` holds 88 house sales in the Boston area. `hprice1$price` is the sale price and `hprice1$assess` is the value the city gave the house for property tax before the sale, both in thousands of dollars. Owners pay tax on the assessed value. A tax consultancy sells appeals to homeowners. It says the city assesses houses above their market value. If the assessments are fair, a house sells on average for its assessed value: the line is price $= 0 + 1 \times$ assess. Fit `reg <- lm(price ~ assess, data = hprice1)`.

a. Report the slope, the intercept and their standard errors.

b. Test $H_0 : \beta_1 = 1$ against $H_A : \beta_1 \neq 1$ at 5%. Why can you not use the t value printed by `summary(reg)` for this test?

c. Test $H_0 : \beta_0 = 0$ against $H_A : \beta_0 \neq 0$ at 5%.

d. Build the 95% confidence intervals for β_1 and for β_0. Do they agree with (b) and (c)?

e. A typical house in the data is assessed at 300 thousand dollars. Build the 95% confidence interval for the mean sale price of houses assessed at 300. If the assessments are fair, this mean is 300. Is 300 inside the interval?

f. Compare the standard error of the intercept with the standard error of the mean price at 300. Why is one so much larger? Use `min(hprice1$assess)`. Should the consultancy's clients trust (b)–(c) or (e)?

Exercise | Start Over | Show Hint | Run Code

```
1 n <- 88
```

Exam practice 17 — Pricing a new route

[#17 · core]

`airfare` holds US airline routes from 1997 to 2000. `airfare$fare` is the average one-way fare in dollars and `airfare$dist` is the distance of the route in miles. Keep the year 2000 (1,149 routes): `a <- airfare[airfare$year == 2000, ]`. Your airline will open a new route of 1,500 miles. Fit `reg <- lm(fare ~ dist, data = a)`.

a. Interpret the slope in cents per mile. Interpret the intercept as the part of a fare that does not depend on distance.

b. Predict the fare for a 1,500-mile route. Build the 95% confidence interval for the mean fare of all 1,500-mile routes from the formula sheet. You need s, $\bar{x}$ and S_{xx}.

c. Build the 95% prediction interval for the fare on your one new route. Check both intervals with `predict()`.

d. The pricing team must set the fare for your new route. An industry report wants the typical fare of 1,500-mile routes. Which interval does each of them need? How many times wider is the prediction interval?

e. The airline also looks at a 4,000-mile route. Why should it not price that route with this regression? Use `max(a$dist)`.

Exercise | Start Over | Show Hint | Run Code

```
1 a <- airfare[airfare$year == 2000, ]
2 n <- 1149
3 x0 <- 1500
```

📝 Exam practice 18 — Does one more letter pay for itself?

[#18 · core]

`charity` holds 4,268 past donors of a Dutch charity. `charity$mailsyear` is the number of letters asking for money that the charity sends the donor each year. `charity$gift` is what the donor gave after the latest letter (0 if nothing). Money is in guilders, the Dutch currency before the euro. Each letter costs the charity 2 guilders: printing, postage and staff time. The fundraising team will send one more letter a year only if it brings in more than it costs. Fit `reg <- lm(gift ~ mailsyear, data = charity)`.

a. Interpret the slope in guilders per extra letter a year.

b. Test $H_0 : \beta_1 \leq 2$ against $H_A : \beta_1 > 2$ at 5%. Report the t statistic, the critical value and the p-value. Does one more letter pay for itself?

c. The board wants to be more careful and uses 1%. Does the decision change?

d. The slope compares donors who already get different numbers of letters. Give one reason why one more letter to the same donor may bring in less than the slope says.

Exercise | Start Over | Show Hint | Run Code

```
1 n <- 4268
```

Exam practice 19 — A realtor with a narrow market

[#19 · stretch]

Back to `houseprices` (Exam practice 10). A realtor in Ames sells only mid-size homes, from 1,200 to 1,800 square feet. She wants the price of one extra square foot from those homes only (444 houses): `mid <- houseprices[houseprices$grlivarea >= 1200 & houseprices$grlivarea <= 1800, ]`.

a. Regress `saleprice` on `grlivarea` twice: with all 973 houses and with the 444 houses in `mid`. Report the slope and its standard error for each. How many times larger is the realtor's standard error?

b. For each fit, compute s, S_{xx} and the standard deviation of `grlivarea`. Three things make a slope more precise: more data, less scatter around the line, and more spread in x. Which one explains most of the change in (a)? Use $SE(\hat{\beta}_1) = s/\sqrt{S_{xx}}$.

c. Build both 95% confidence intervals for the slope. A buyer asks: "What is one extra square foot worth?" What range can each fit give her?

d. R^2 falls from 0.61 to 0.19. Does that mean size matters less for mid-size homes? Compare the two slopes.

Exercise | Start Over | Show Hint | Run Code

```
1 mid <- houseprices[houseprices$grlivarea >= 1200 & houseprices$grlivarea <= 1800, ]
2 n_all <- 973
3 n_mid <- 444
```

📝 Exam practice 20 — More data: what gets more precise?

[#20 · stretch]

`h` is the sample of 200 ENIGH households from Exam practice 12, and `H` holds all 74,639 households with positive income. Both are already loaded, with `food_spending` and `monthly_income` in pesos a month. A supermarket chain plans its stores with a rule: households spend 10 centavos of every extra peso of income on food, so $\beta_1 = 0.10$ in the regression of `food_spending` on `monthly_income`.

a. Fit the regression with `h`. Test $H_0 : \beta_1 = 0.10$ against $H_A : \beta_1 \neq 0.10$ at 5%.

b. Do the same test with `H`.

c. How many times smaller is the standard error of the slope with `H`? Why can the chain's rule pass one test and fail the other?

d. For a household with an income of 20,000 pesos a month, get the 95% confidence interval for the mean food spending and the 95% prediction interval, from both fits. Which interval became much narrower with more data? Which one hardly changed? Use the formula sheet to explain why.

e. The chain needs two forecasts for households earning 20,000 pesos: the average food spending of the 3,000 such households near a new store, and the food spending of one loyalty-card customer. For which forecast does a bigger survey help?

Exercise | Start Over | Show Hint | Run Code

```
1
```

Appendix — optional

The algebra behind the results in the main flow.

Appendix — what “linear” means here

A linear estimator is a weighted sum of the y_i:

$$\tilde{\beta}_1 = \sum_i c_i \, y_i$$

OLS is one of them. Since $\sum (x_i - \bar{x})\bar{y} = 0$:

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\sum (x_i - \bar{x})\, y_i}{\sum (x_i - \bar{x})^2} \qquad \text{so} \qquad c_i^{OLS} = \frac{x_i - \bar{x}}{S_{xx}}$$

Points far from $\bar{x}$ get the biggest weights. That is the plank from the slider slide again.

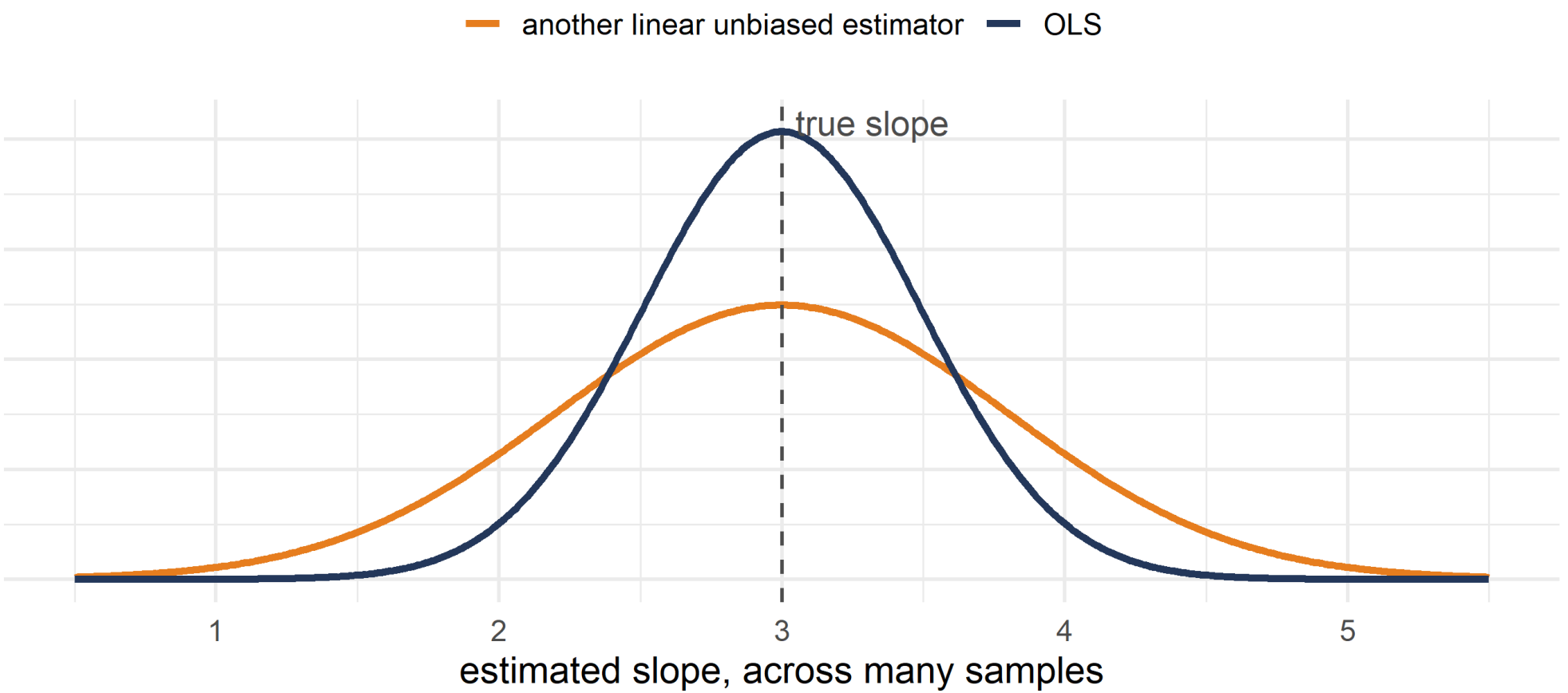
← Back to the Gauss–Markov theorem

Appendix — an update from 2022

- Science is still moving on this theorem.
- A 2022 paper by Bruce Hansen shows that the word “linear” is not needed.
- Under our assumptions, OLS is the best unbiased estimator of any kind: BUE, not only BLUE.

← Back to the Gauss–Markov theorem

Appendix — the theorem in a picture



- Both curves are centred on the true slope: both estimators are unbiased.
- The OLS curve is the narrowest. In any sample, OLS is the least likely to land far from the truth.
- Gauss–Markov says no linear unbiased estimator has a narrower curve than OLS.

← Back to Which one moves more?

Appendix — changing the units, again

Last week: rescale x to $z = ax + c$ and the slope becomes $\hat{\beta}_1 / a$.

A classic version: a house-price regression is estimated on lot area in square feet, and you want to use it in a country that uses square metres. How do $\hat{\beta}_1$ and $SE(\hat{\beta}_1)$ change?

The same question for your client: her investors are in Texas and think in Fahrenheit, $F = 1.8\,C + 32$.

🖊️ Appendix exercise — Trips per degree Fahrenheit

Make `bikes$temp_f`, the temperature in Fahrenheit: $1.8 \times$ `bikes$temp` $+32$.

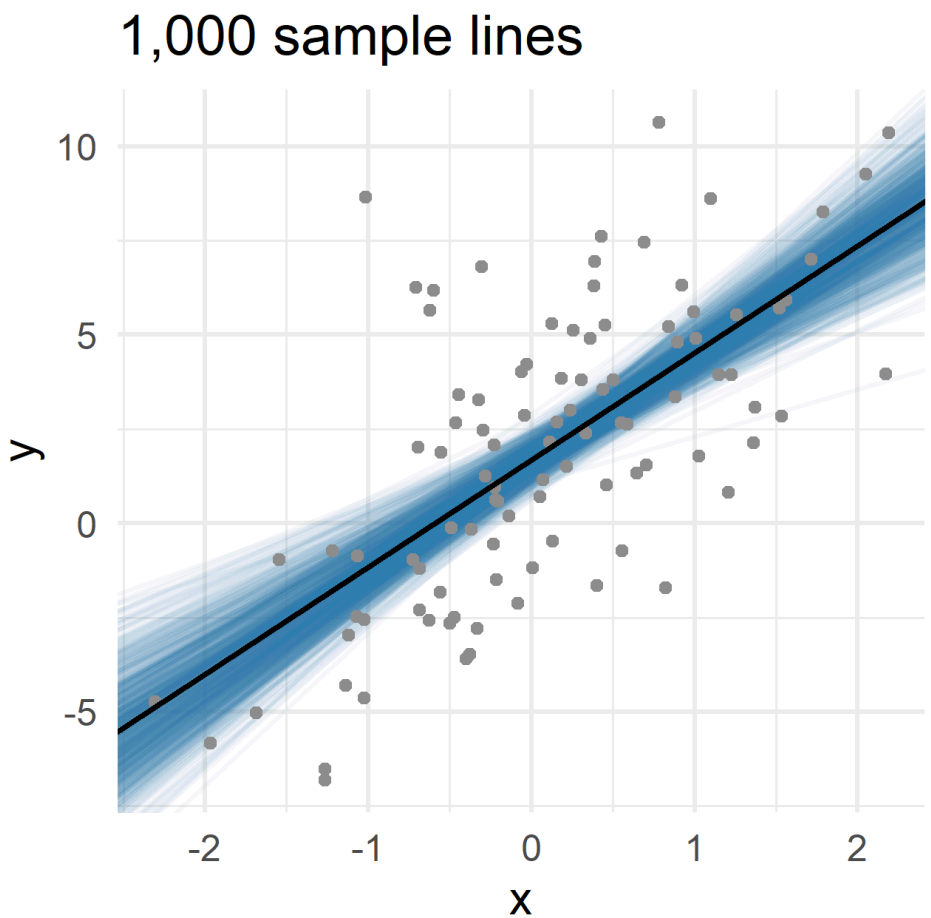
1. Predict first: what will the new slope be? The new standard error of the slope?
2. Fit `trips` on `temp_f` and get the slope and its standard error from `vcov()`.
3. Divide the slope by its standard error, in both regressions. What do you notice?

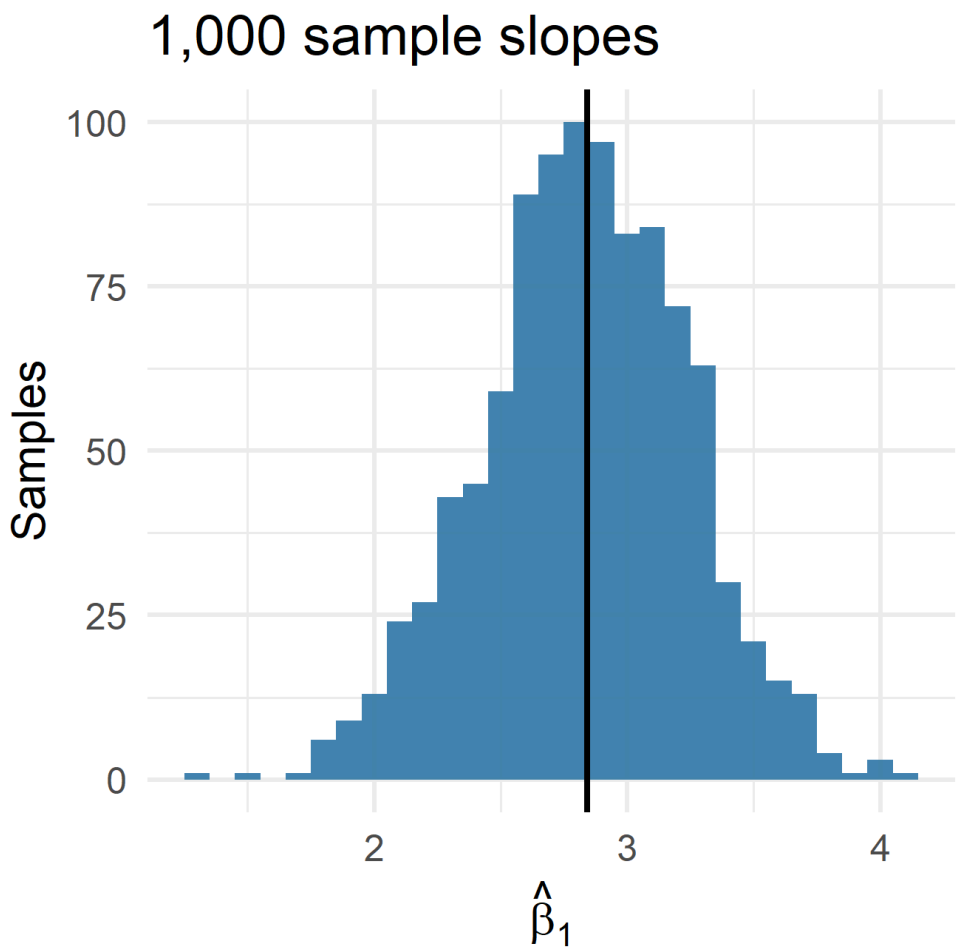
Exercise | Start Over | Show Hint | Run Code

```
1
```

← Back to What the standard error says

Appendix — 1,000 samples from the population of 100





- Left: each blue line is one sample's fit. Black: the population line.
- Right: the 1,000 slopes pile up around the population slope, in a bell shape.

The 1,000 slopes average 2.83, against a population slope of 2.84. Their standard deviation is 0.41: a typical sample slope misses by about that much.

← Back to the sampler

Appendix — the question on the table

Last week you told your Ecobici client:

> A one-degree warmer day brings about 724 more trips.

She is about to order new bikes against that number. Her finance director asks two questions:

1. If we had collected three different years of data, would we still get 724?
2. Could the true effect be zero, and 724 just luck?

Both are questions about uncertainty. Answering them is this week.

Appendix — the data, one more column

`bikes` — the same 781 weekdays in Mexico City, 2017–2019.

Column	What it is
`trips`	Ecobici trips that day
`temp`	average temperature, °C
`pm25`	fine-particle pollution, µg/m^3
`friday`	1 if Friday, 0 otherwise
`year`	2017, 2018 or 2019

Each year has about 260 weekdays. Each year is a sample of its own.

Appendix exercise — Three years, three slopes

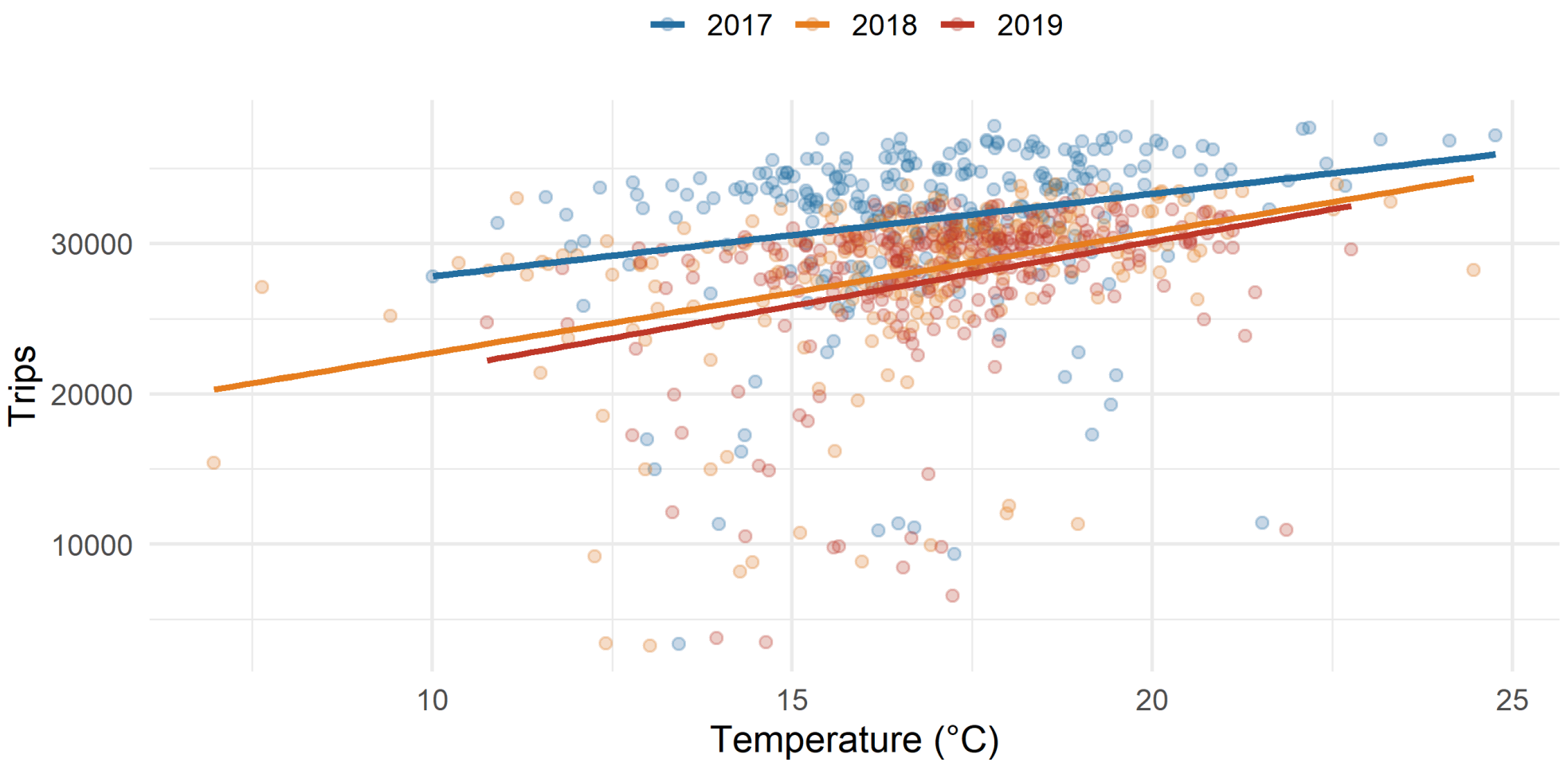
Fit the regression of `bikes$trips` on `bikes$temp` three times: once with only the 2017 days, once with 2018, once with 2019.

1. What is the slope in each year?
2. Which year gives the largest slope? Which gives the smallest?
3. Your client wants to order bikes using one year of data only. How many trips per degree would she plan for if she picked the smallest year? The largest?

Exercise | Start Over | Show Hint | Run Code

```
1
```

Appendix — three years, three lines



- Same city, same bikes, three samples — three different slopes.
- The slope we estimate depends on which days happened to be in the sample.

← Back to What the standard error says

Appendix — the variance of the slope

Treat the x_i as fixed numbers. Only the errors u_i are random.

$$\mathrm{Var}(\hat{\beta}_1) = \mathrm{Var}\left(\frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{S_{xx}}\right)$$

$$= \mathrm{Var}\left(\sum_i \frac{x_i - \bar{x}}{S_{xx}}\, y_i\right)$$

$$= \mathrm{Var}\left(\sum_i \frac{x_i - \bar{x}}{S_{xx}}\, u_i\right)$$

$$= \sum_i \mathrm{Var}\left(\frac{x_i - \bar{x}}{S_{xx}}\, u_i\right)$$

$$= \sum_i \left(\frac{x_i - \bar{x}}{S_{xx}}\right)^2 \mathrm{Var}(u_i)$$

$$= \sigma^2\, \frac{\sum(x_i - \bar{x})^2}{S_{xx}^2} = \sigma^2\, \frac{S_{xx}}{S_{xx}^2} = \boxed{\frac{\sigma^2}{S_{xx}}}$$

Start
The OLS slope, with $S_{xx} = \sum(x_i - \bar{x})^2$ — a fixed number.

Why $\bar{y}$ drops out
$\sum(x_i - \bar{x})(y_i - \bar{y})$
$= \sum(x_i - \bar{x})\, y_i - \bar{y} \sum(x_i - \bar{x})$
The red part is zero: $\sum(x_i - \bar{x}) = n\bar{x} - n\bar{x} = 0$.

Only the error is random
Plug in $y_i = \beta_0 + \beta_1 x_i + u_i$; the red part is fixed.
$\mathrm{Var}(c + Z) = \mathrm{Var}(Z)$ for a fixed number c.

Independence: the sum comes out
$\mathrm{Var}(A + B) = \mathrm{Var}(A) + \mathrm{Var}(B) + 2\,\mathrm{Cov}(A, B)$
The u_i are independent, so every covariance is zero.

Fixed numbers come out squared
$\mathrm{Var}(a\,Y) = a^2\,\mathrm{Var}(Y)$, here with the fixed number $a = (x_i - \bar{x})/S_{xx}$.

Homoskedasticity: σ^2 comes out
$\mathrm{Var}(u_i) = \sigma^2$ for every i: one number for everyone, so it leaves the sum.
And $\sum(x_i - \bar{x})^2 = S_{xx}$, so one S_{xx} cancels.

← Back to The variance of the slope

Appendix — the variance of the intercept

Treat the x_i as fixed numbers. Only the errors u_i are random.

$$\mathrm{Var}(\hat{\beta}_0) = \mathrm{Var}\left(\bar{y} - \hat{\beta}_1 \bar{x}\right)$$

$$= \mathrm{Var}\left(\bar{u} - \hat{\beta}_1 \bar{x}\right)$$

$$= \mathrm{Var}(\bar{u}) + \bar{x}^2\,\mathrm{Var}(\hat{\beta}_1) - 2\bar{x}\,\mathrm{Cov}(\bar{u}, \hat{\beta}_1)$$

$$= \mathrm{Var}(\bar{u}) + \bar{x}^2\,\mathrm{Var}(\hat{\beta}_1)$$

$$= \frac{\sigma^2}{\cancel{n}} + \bar{x}^2 \frac{\sigma^2}{S_{xx}} = \boxed{\sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)}$$

Start
The OLS intercept. $\bar{x}$ is a fixed number.

Only the error is random
Average the model over the n days: $\bar{y} = \beta_0 + \beta_1\bar{x} + \bar{u}$; the red part is fixed.
$\mathrm{Var}(c + Z) = \mathrm{Var}(Z)$ for a fixed number c.

Variance of a difference
$\mathrm{Var}(A - b\,B) = \mathrm{Var}(A) + b^2\,\mathrm{Var}(B) - 2b\,\mathrm{Cov}(A, B)$, here with the fixed number $b = \bar{x}$.

Independence: the covariance is zero
From the previous slide, $\hat{\beta}_1$ is a fixed number plus $\sum_i \frac{x_i - \bar{x}}{S_{xx}} u_i$, and $\bar{u} = \sum_i \frac{1}{n} u_i$.
The u_i are independent, so only same-day pairs count:
$\mathrm{Cov}(\bar{u}, \hat{\beta}_1) = \frac{\sigma^2}{nS_{xx}} \sum(x_i - \bar{x}) = 0$.

Homoskedasticity: σ^2 comes out
$\mathrm{Var}(\bar{u}) = \sigma^2/n$: the average of n independent errors, each with variance σ^2.
$\mathrm{Var}(\hat{\beta}_1) = \sigma^2/S_{xx}$ from the previous slide.

← Back to The variance of the slope

Appendix — why the slope is normal in big samples

The OLS slope is a linear combination of the errors:

$$\hat{\beta}_1 = \beta_1 + \sum_i \underbrace{\frac{x_i - \bar{x}}{S_{xx}}}_{\text{weight}} u_i$$

Call each weighted error $Z_i = \frac{x_i - \bar{x}}{S_{xx}} u_i$, so that $\hat{\beta}_1 - \beta_1 = \sum_i Z_i$.

By the Central Limit Theorem, a sum of many independent, mean-zero terms is approximately normal:

$$\sum_i Z_i \xrightarrow{d} N\left(0,\ \mathrm{Var}\left(\sum_i Z_i\right)\right)$$

And

$$\mathrm{Var}\left(\sum_i Z_i\right) = \sum_i \mathrm{Var}(Z_i) = \frac{\sum_i (x_i - \bar{x})^2}{S_{xx}^2}\,\sigma^2 = \frac{S_{xx}}{S_{xx}^2}\,\sigma^2 = \frac{\sigma^2}{S_{xx}}$$

so

$$\hat{\beta}_1 - \beta_1 \xrightarrow{d} N\left(0,\ \frac{\sigma^2}{S_{xx}}\right)$$

← Back to Big samples: the CLT

Appendix — testing the intercept, and one-sided tests

Two cases used in Exercise 5.

- The intercept is tested exactly like the slope, with $SE(\hat{\beta}_0)$. You built it in Exercise 1; `vcov(reg)[1, 1]^0.5` gives it too.
- Sometimes only one direction has a business cost. Then the alternative is one-sided, and so is the p-value:

Alternative	Reject when	p-value in R
$H_A : \beta \neq \beta^0$	$\lvert T \rvert > t_{n-2,\,1-\alpha/2}$	`2 * (1 - pt(t_stat, df))` if $T > 0$, `2 * pt(t_stat, df)` if $T < 0$
$H_A : \beta > \beta^0$	$T > t_{n-2,\,1-\alpha}$	`1 - pt(t_stat, df)`
$H_A : \beta < \beta^0$	$T < -t_{n-2,\,1-\alpha}$	`pt(t_stat, df)`

← Back to Exercise 5

Appendix — testing against a number that is not zero

The bike maker's planning guide says each extra degree brings 1,000 more trips. The guide was built on data from another city. Your client will order bikes by this guide unless the Mexico City data disagree.

$$H_0 : \beta_1 = 1{,}000 \qquad \text{against} \qquad H_A : \beta_1 \neq 1{,}000$$

Two-sided, because both mistakes cost money. If the true effect is smaller, she buys bikes that stand unused. If it is larger, stations run empty on hot days.

← Back to Exercise 6

Appendix exercise — The agent's 200 pesos

Back to the landlord from last week. Adding 10 m^2 to an apartment costs 150,000 pesos. A real-estate agent tells her that each m^2 adds 200 pesos a month in rent, so the work pays back in 75 months.

`rents` has 150 apartments: `rents$size` (m^2) and `rents$rent` (pesos a month).

1. Test $H_0 : \beta_1 = 200$ against $H_A : \beta_1 \neq 200$ at 5%.
2. Is the agent's claim consistent with her data?
3. What does that mean for the 75-month payback?

Exercise | Start Over | Show Hint | Run Code

```
1 n <- 150
```

← Back to Exercise 6

Appendix — why the error cancels in the average but not on one day

On any day at temperature x_0, trips are $y = \beta_0 + \beta_1 x_0 + u$, with $E(u) = 0$ and $\mathrm{Var}(u) = \sigma^2$, independently from day to day.

The average of m days at x_0:

$$\bar{y}_m = \frac{1}{m}\sum_{j=1}^{m}(\beta_0 + \beta_1 x_0 + u_j) = \beta_0 + \beta_1 x_0 + \bar{u}_m$$

$$E(\bar{u}_m) = 0, \qquad \mathrm{Var}(\bar{u}_m) = \frac{\sigma^2}{m} \longrightarrow 0 \quad \text{as } m \to \infty$$

So the average of many 22 °C days is $\beta_0 + \beta_1 x_0$ itself, with no error left. The interval only has to cover the error in our estimate of the line: $\mathrm{Var}\left(\hat{\beta}_0 + \hat{\beta}_1 x_0\right) = \sigma^2\left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)$.

← Back to The mean response

Appendix — one day keeps its error

Tomorrow is one day: $y_0 = \beta_0 + \beta_1 x_0 + u_0$. Our forecast is $\hat{y}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0$, so the forecast error is

$$y_0 - \hat{y}(x_0) = \left(\beta_0 + \beta_1 x_0 - \hat{\beta}_0 - \hat{\beta}_1 x_0\right) + u_0$$

u_0 belongs to tomorrow, so it is independent of the 781 days used to fit the line. The two variances add:

$$\operatorname{Var}\left(y_0 - \hat{y}(x_0)\right) = \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right) + \sigma^2 = \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)$$

The extra σ^2 does not shrink with more data. A bigger sample pins down the line, not tomorrow's surprise.

← Back to The mean response

Appendix exercise — The days near 22 °C

There are 19 days in `bikes` with `bikes$temp` between 21 and 23 °C.

1. What are the lowest and the highest number of trips on those days?
2. What is their mean, and their standard deviation?
3. What is the standard error of their mean? Compare it with the standard deviation.

Exercise | Start Over | Show Hint | Run Code

```
1
```

← Back to What the 22 °C days show